2005:7

Ari Kokko, Thomas Mathä
and Patrik Gustavsson Tingvall

European Integration and Trade Diversion:

Yeats revisited

Sieps
Swedish Institute for European Policy Studies

Ari Kokko, Thomas Mathä
and Patrik Gustavsson Tingvall

European Integration and Trade Diversion:

Yeats revisited

– Sieps 2005:7 –

Report No. 7
October/2005

Publisher: Swedish Institute for European Policy Studies.

The report is available at
www.sieps.se

The report can also be ordered from
info@sieps.se

The opinions expressed in this report are
those of the author and are not necessarily
shared by SIEPS.

Cover: Svensk Information AB

Print: EO-print AB

Stockholm, October 2005

ISSN 1651-8942
ISBN 91-85129-30-5

FOREWORD

The Swedish Institute for European Policy Studies, SIEPS, conducts and promotes research on – and analysis of – European policy issues, within the following disciplines: political science, law and economics and trade. SIEPS considers it important to broaden and intensify research into matters that are significant for the future development of the European Union and, towards that end, actively seeks to develop close cooperation with institutes and research centres in Europe and globally. SIEPS strives to act as a link between the academic world and policy-makers at various levels. By publishing reports and arranging seminars and conferences SIEPS aims to further stimulate research on the future of Europe.

The last decades have been characterized by deep regional integration in Europe as well as the emergence of new integration agreements in other parts of the world. This process has spurred an extensive debate about what the economic effects of integration are, both on insiders and outsiders. However, in many cases, the results are ambiguous, theoretically as well as empirically. The purpose of the present study is to contribute to the understanding of the links between regional integration and trade patterns by looking at the trade effects of the European integration. The report highlights the consequences for outsiders in general and developing countries in particular.

Stockholm, October 2005
Annika Ström Melin
Director

ABOUT THE AUTHORS

Ari Kokko is Professor at the European Institute of Japanese Studies, Stockholm School of Economics. Patrik Gustavsson is Assistant Professor at the same Institute. In addition to studies focusing on Asian development, they have also worked on various issues related to the Swedish and European economies. An earlier Sieps publication is *Regional integration och regionala handelsavtal* (2004), which provides an overview of recent theoretical and empirical research on regional integration.

Thomas Mathä is Assistant Professor at the Central Bank of Luxembourg. His work is concentrated on research on European integration with a focus on trade and monetary issues.

TABLE OF CONTENTS

1 INTRODUCTION 6

2 BACKGROUND: TRADE DIVERSION À LA YEATS (1998) 8

3 EUROPEAN INTEGRATION AND TRADE 12
3.1 European trade structure 15
3.2 EU trade with transition economies and developing countries 22

4 COMPARATIVE ADVANTAGE, REGIONAL ORIENTATION, AND TRADE DIVERSION 29
4.1 What is the relation between ROI and A-RCA? 42
4.2 How do these results compare with other studies? 43

5 CONCLUSIONS 48

6 REFERENCES 50

APPENDIX TABLE 1. COUNTRY GROUPS 54

APPENDIX TABLE 2. INDUSTRY CATEGORIES 56

SAMMANFATTNING PÅ SVENSKA 59

1 INTRODUCTION

Regional integration has arguably been the most important feature of European economic development for several decades. Starting with the Coal and Steel Union and the Treaty of Rome in the 1950s, the integration process has gradually extended to include new member countries and new policy areas. The most significant developments during the last few years are the establishment of the European Monetary Union, with 12 member countries adopting a common currency in 2001, and the entry of 10 new member countries in May 2004. These developments make the European Union the world's most comprehensive regional integration agreement, even before the ratification of the new EU constitution, which was signed by the heads of state of the 25 member countries in October 2004.

Since the early 1980s, regional integration has also become a global phenomenon. The number of regional integration agreements (RIAs) has increased rapidly, and it is now difficult to identify any country that does not participate in some kind of regional integration arrangement. Well over one-third of world trade takes place within different kinds of trade blocs. Yet, despite the proliferation of RIAs, the debate about the costs and benefits of regionalism continues to be lively. One of the main questions is whether regional agreements serve as an obstacle or a "stepping stone" in the process towards global free trade (World Bank 2000). The opponents see RIAs as fostering protectionism, encouraging governments to look inward rather than outward, which undermines the multilateral trading system. Others maintain that regional economic groupings have not impeded the development of freer world trade and have sometimes helped to promote it. One reason for the diversity of views is that the economic effects of customs unions and other regional integration schemes are not clear-cut. One concern is that the new intraregional trade created by RIAs may come at the expense of trade relations with the rest of the world – diverting trade from more efficient producers – which reduces the welfare of third countries and has ambiguous effects on the welfare of the RIA members.

This debate has been particularly intensive in South America, where MERCOSUR, a customs union including Argentina, Brazil, Paraguay, and Uruguay, was established in 1991. This RIA, covering about 70% of Latin America's land surface and 50 % of the continent's population, had an immediate and dramatic effect on regional trade and investment. Within four years, the value of trade among the member countries tripled, the inflows of FDI increased dramatically, and the agreement was widely advocated by the governments of the member countries. This notwithstanding, there were critical views regarding the outcomes of the trade expansion. For in-

stance, Alexander Yeats, the chief economist of the International Trade Division of the World Bank, argued in an unpublished report that leaked to the press that "discriminatory barriers" towards outsiders had induced consumers in the region to buy expensively produced (tariff-free) manufactured goods from their neighbors rather than cheaper (taxed) goods from other countries. In other words, the report asserted that MERCOSUR diverted trade and reduced the welfare of both members and non-members. The initial version of the report concluded that these findings "appear to constitute the most convincing, and disturbing, evidence produced thus far concerning the potential adverse effects of regional trade arrangements".[1] While the version that was published two years later (Yeats 1998) used milder phrases, it still made a strong case against MERCOSUR.

The aim of this paper is to re-examine the relation between regional integration and trade, by looking at the trade effects of European integration. Departing from the framework set up by Yeats (1998), we explore the development of Europe's regional and external trade since the early 1960s. Unlike Yeats, who only looked at the development of MERCOSUR exports, we also discuss the import side. Our main conclusion is that the signs of trade diversion are much weaker than what Yeats (1998) presented: in particular, we find no aggregate evidence that European integration has been detrimental to outsiders. Another central conclusion concerns differences over time. The deepening integration process in Europe has coincided with changes in the trade effects of integration, so that the risks for trade diversion seem to be diminishing over time.

The remainder of this paper is organized as follows. Section 2 summarizes Yeats' (1998) framework, section 3 overviews the main stages of European integration and provides some descriptive statistics for EU trade, section 4 applies a version of the Yeats (1998) methodology to European trade statistics, and section 5 concludes and summarizes the findings.

[1] Quoted in Krugman and Obstfeld, 2000.

2 BACKGROUND: TRADE DIVERSION À LA YEATS (1998)

The standard analysis of the trade effects of regional integration dates back to Viner (1950), Meade (1955), and Lipsey (1957), who distinguished between two major effects from the establishment of customs unions – trade creation and trade diversion. Trade creation refers to the new trade relations that emerge, as trade barriers between RIA members are reduced. It is generally seen as a positive effect, both from the perspective of consumer welfare (since market prices fall) and allocative efficiency (because more efficient producers replace less efficient ones). Trade diversion occurs when the increase in regional trade comes about at the expense of imports from more efficient producers located outside the RIA. The welfare effects of trade diversion may be negative, since less efficient regional producers capture market shares from more efficient outsiders: the possible gains in consumer welfare and producer surplus within the RIA must be weighed against the losses made by the outside producers. Both trade creation and trade diversion are likely to occur in every RIA, and the aggregate effect therefore depends on their relative magnitudes. Theory does not give any unambiguous answers regarding these magnitudes, so determining the net effect is largely an empirical issue.

It is therefore not surprising that a vast empirical literature on the trade effects of RIAs has emerged during the past decades, as Europe's internal trade barriers have come down and regional agreements have grown more common throughout the world. Most of the recent empirical studies can be categorized in two groups depending on what methodology is used – computable general equilibrium (CGE) models or gravity equations. The CGE models are typically ex ante exercises intended to provide estimates of the expected growth and trade effects of regional integration, and have been employed to explore the effects of European integration (Baldwin *et al.* 1994, Haaland and Norman 1992, Harrison *et al.* 1994, 1997), North American integration (Brown *et al.* 2003), and a host of other RIAs (Harrison *et al.* 2002, 2003, Rutherford *et al*. 1997, Scollay and Gilbert 2000). However, it is often difficult to model regional integration in detail – it is often treated as a homogenous decrease in trade costs for intra-regional trade – and it is hard to differentiate between trade creation and trade diversion in these models. Gravity equations, where dummies for membership in various RIAs are easily added to a host of trade data, are popular for ex post assessments, as they may provide more direct tests of trade effects (see e.g. Adams *et al*. 2003 for a survey). One of their weaknesses is that it is difficult to control for the growth effects of integration: such effects en-

ter as exogenous shifts in other explanatory variables, and may reduce the impact of the variables that distinguish members from non-members. Hence, while each of these methods can provide important insights regarding trade effects, neither of them will reveal the full picture.

To get a more complete understanding of the key relations between variables, it is therefore useful to apply several different methodologies – a form of triangulation – to the same problem. Regarding trade diversion, another alternative approach has recently been proposed by Yeats' (1998), who bases his analysis of the trade and efficiency effects of MERCOSUR on detailed comparisons of changes in two indices – the regional orientation index and the revealed comparative advantage index. The Regional Orientation Index (ROI) measures the relative importance of intra-regional exports for a product or product group. It is defined as

$$\mathbf{ROI_j = (x_{jr}/X_r) / (x_{j0}/X_o)} \qquad \mathbf{(1)}$$

where (x_{jr}/X_r) is the share of product j in intra-regional exports and (x_{j0}/X_o) is its share in exports to third countries. An index value above unity indicates a concentration of exports to the regional markets.

The Revealed Comparative Advantage Index (RCA) is used to measure the competitiveness of individual export goods in a country's or region's trade. The RCA index is defined as

$$\mathbf{RCA_j = (x_{j0}/X_o) / (x^*_{jw}/X^*_w)} \qquad \mathbf{(2)}$$

where (x_{j0}/X_o) is the share of product group j in the country's or region's external trade and (x^*_{jw}/X^*_w) is the country's or region's share of total world exports. The index value exceeds unity for products where the country has a revealed comparative advantage, in the sense that the export share for the product is higher than the country's overall weight in world trade. To adapt the measure to an analysis of RTAs, it is necessary to exclude intra-regional trade from both terms in the denominator, to be sure that internal preferences guaranteed to the members of the union do not bias the result.

While neither of these indices provides much useful information when observed at a single point in time, their development over time may reveal interesting information about changes in trade patterns. In particular, Yeats (1998) argued that a pattern where large increases in ROI coincide with weakening RCA would constitute evidence of trade diversion from more efficient external suppliers to less efficient intra-regional producers. This was also the finding from Yeats' analysis of MERCOSUR trade during the period 1988 – 1994. Intra-MERCOSUR trade increased mainly in product

groups with weak external comparative advantages, and the largest increases in the ROI index were actually recorded for products with falling RCA. This, in his interpretation, implied a direct cost in welfare terms, since "to the extent that non-members' exports are restricted, discriminatory trade barriers would deny consumers access to higher-quality and lower-priced goods in MERCOSUR's internal markets" (Yeats 1998: 26).

The reason for this inefficiency, according to Yeats, was MERCOSUR's discriminatory trade policy, induced by high internal preference margins and non-tariff barriers. The counterfactual comparison he had in mind was with liberalization on a non-discriminatory basis, which would arguably not have resulted in any shift towards regional producers. He therefore concluded that "…[the results] constitute convincing evidence that regional preferences can affect trading partners strongly and detrimentally for both member and non-member countries" (Yeats 1998: 25).

After having leaked to both *the Wall Street Journal* and *the Financial Times,* the unpublished version of Yeats' study met heavy criticisms from the proponents of regional integration in general and MERCOSUR in particular. Some critics pointed out that Yeats paid insufficient attention to the protection structure of MERCOSUR and the external trade liberalization pursued by the countries between 1988 and 1994, which resulted in significant increases in imports from third countries after 1994 (Mendoza and Hudgins 1997, Devlin 1996). Cohen and Nagarajan (1998) noted that agricultural goods and other primary products were excluded from the analysis, making it hard to generalize the conclusions to the overall trade performance of the region, where primary products are still important. External reasons for shifting exports to regional partners were not taken into account – for instance, the fall of communism in Eastern Europe may have partially shifted EU's import demand from Latin America to the Central and East European transition economies (Hudgins 1997). The focus on exports alone was also considered too narrow: an analysis of exports as well as imports would have provided a more complete picture of changes in trade patterns (Sepulveda and Aguirre 1997). Accordingly, Devlin (1996) and Nagarajan (1998) asserted that there might not be any clear evidence of trade diversion when the structure of imports is included in the analysis. Another limitation of Yeats' analysis is that he focused on the static effects of regional integration. By focusing on trade shares, Yeats neglected the dynamic effects of trade – the increases in overall trade that may have occurred as a result of regional integration, providing a larger market for domestic as well as foreign producers. Perhaps most importantly, Yeats' choice of industries for the analysis was challenged. By selecting the 30

three-digit industries with the largest relative increase in regional orientation, Yeats (1998) included several industries that were relatively unimportant for the overall volumes of trade. Bercovica *et al*. (2003) showed that there are no signs of trade diversion if the set of industries is selected on the basis of the largest absolute increases in regional trade. Hence, it is not obvious that the worries regarding trade diversion are well founded in the case of MERCOSUR, and it is not clear what the pattern is for other integration agreements, like the EU.

3 EUROPEAN INTEGRATION AND TRADE

Just like international trade in general, European trade has expanded rapidly during the last 50 years. In addition to falling trade and information costs and the positive effects of multilateral trade liberalization through the GATT (and WTO since 1995), Europe has also benefited from a sequence of regional integration agreements since the early 1950s. As a result, the number of countries participating in Europe's core integration agreements has increased from 6 in the 1950s to 25 in 2004. The first substantial RIA in the post-war period was the European Coal and Steel Community, established by Belgium, France, Germany, Italy, Luxembourg, and the Netherlands in 1951 through the Treaty of Paris. The purpose of this agreement was to integrate the member states' markets for coal and steel, not only for reasons of economic efficiency but perhaps mainly as a way to maintain peaceful relations between the members: one of the main arguments for integration was that closely integrated nations would not start wars with each other. In 1957, the Treaty of Rome added two more Communities, the European Atomic Energy Community (EURATOM) and the European Economic Community (EEC). These three were eventually merged under the name European Communities (EC) in 1967. The EEC was essentially a customs union (completed in 1968) that abolished tariffs between the member states and introduced a common external trade policy, but the presence of other community institutions contributed to a deeper integration than in most other customs unions. For instance, the European Court, with force of law in member states, was already established in connection with the Treaty of Rome, and a common agricultural policy emerged after 1962.

The first enlargement of the EC took place in 1973, when Denmark, Ireland and the United Kingdom joined the community. The enlargement of the EC coincided with a turbulent period in European development, with the collapse of the Bretton-Woods system of fixed exchange rates adding to the uncertainty created by several oil price shocks. The EC attempted to handle this uncertainty through the introduction of several measures to promote monetary integration, leading to the establishment of the European Monetary System (EMS) in 1979. This construction aimed to establish fixed exchange rates between the member states, but it also entailed some coordination of fiscal policy to maintain these fixed rates. There were also attempts to deepen the degree of trade integration, but with limited success. The main trade related achievement was instead a bilateral free trade agreement with the European Free Trade Area (EFTA) in the mid-1970s. The enlargement of the EC also proceeded in the 1980s, when Greece (1981) and Portugal and Spain (1986) joined the community. By

that time, the EC-EFTA free trade area covered 18 countries: apart from the twelve EC countries, these were Iceland, Norway, Sweden, Finland, Austria, and Switzerland.

The mid-1980s also saw a shift in the character of the European trade integration process. The achievements during the period from the late 1950s to the mid-1980s were mainly related to the abolition of internal tariffs (which was largely done by 1968) and the reduction of external tariffs through the GATT and bilateral agreements, e.g. with the EFTA countries. Yet, there was a general feeling that the integration process had come to a standstill towards the mid-1980s, as economic growth in the EC fell behind that in the US and Japan. To revive the stagnant European economy, the EC therefore launched the so called Single Market Program (SMP), which was intended to be completed by 1992.

The aim of the SMP was to dismantle the numerous non-tariff barriers that had been created over the preceding decades and that effectively prevented the completion of the internal market. The ultimate objective was to finally achieve the four freedoms already set out in the Treaty of Rome: free movement of goods, labour, capital and services. The Emerson *et al*. (1988) report, which assessed the economic impacts and gains of deeper integration on behalf of the European Commission, grouped the remaining barriers into five categories: tariffs, quotas, cost increasing barriers, market-entry restrictions, and market-distorting subsidies and practices. The abolition of the few remaining tariffs and quotas was technically uncomplicated, but also of relatively small importance. Most of the economic gains were instead expected to emanate from dismantling the remaining cost-increasing barriers, (e.g. delays due to customs procedures and different national regulations), market entry restrictions, (e.g. government procurement restrictions, restrictions on the right of establishment of various service industries and professions, and restrictions on the entry into some regulated markets), and from restricting market-distorting subsidies and practices (e.g. collusion and abusive use of dominant positions of private enterprises). These gains were mainly expected to emanate from better exploitation of economies of scale and tougher competition among member countries: unlike earlier stages of regional integration, this "new" form of integration was also expected to improve the competitiveness of European producers in comparison with producers from other countries, e.g. the US and Japan (see Gustavsson and Kokko 2004).

The announcement of the SMP not only affected the EC (which now became known as the European Union), but it also had effects outside Europe. Many of Europe's trade partners voiced fears about the emergence

of a "Fortress Europe" where internal trade would expand at the expense of trade with the rest of the world. As a result, by the early 1990s, all EFTA countries except Iceland had announced their intention of joining the EU in the future. The first step in this expansion was the establishment of the European Economic Area (EEA) agreement in 1994, which essentially extended the Single Market to include the EFTA countries (although Switzerland opted not to join). In 1995, Austria, Finland and Sweden joined the European Union, but Norway decided to stay outside after a no-vote in a referendum. This was the second "no" in a Norwegian referendum: Norway could have acceded to the EC already in 1973, together with the UK, Ireland, and Denmark, but popular opinion was opposed to membership.

Simultaneously, the EU had taken further steps to deepen and widen the integration process. The Maastricht Treaty from 1992 set the aim to increase the degree of political integration in Europe and outlined the establishment of a European Monetary Union (EMU) by 1999, as well as the introduction of a common currency by 2002. Following the fall of the Iron Curtain and the Soviet Union, the former socialist countries in Central and Eastern Europe were looking towards western Europe for support and partnership. The EU recognized the need to contribute to the integration of these countries, and promised membership in the Union if they could meet the so called Copenhagen Criteria from 1993, which called for stability, democracy, rule of law, human rights and respect for minorities, market development, and convergence of national regulations towards EU law. These requirements were subsequently formalized in Association Agreements outlining the convergence process. Ten Central and Eastern European countries – Estonia, Latvia, Lithuania, Poland, Hungary, the Czech Republic, Slovakia, Slovenia, Malta and Cyprus – eventually joined the EU in 2004.

In the subsequent discussion and analysis, we will distinguish between three broad phases in the process of European integration (although we will use the term EU for all three phases). The first phase is made up of the period up to 1973. This is characterized by the establishment of the EEC, with the customs union at the center. It can be seen as an example of "traditional" regional integration, where the main effects stem from the reduction of formal tariff barriers between the member states. The second phase runs from 1974 to 1986, and involves consolidation of the customs union agreement together with the first stages of monetary integration. The third stage, from 1987 to the present (or, in our data analysis, to 2000), includes a deepening trade integration process leading to the Single Market, as well as deeper integration in the monetary area, through the EMU. This

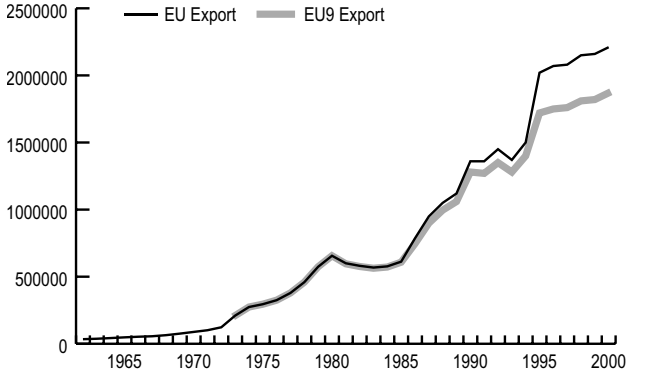
process can be characterized as an example of "new" integration, where the aim is to create a truly single market through the removal of non-tariff barriers and the harmonization (or more often, the mutual recognition) of product standards and other rules and regulations with an impact on trade. A fourth phase of integration, which is not covered by our analysis, could be illustrated by the enlarged EU that emerged after the accession of ten new member countries in 2004. This enlargement not only involved a larger number of countries than earlier enlargements, but it also differed in character, since the income levels and economic structures of the new members were notably different from those of the 15 incumbents.

3.1 European trade structure

The structure and volume of European trade has changed substantially during the past four decades, as European integration has become progressively deeper and more countries have joined the common market. Figure 1 summarizes the growth of EU's total exports as well as the total exports of a group labelled EU9 during the period 1962-2000. While the composition of the EU group has changed over time with the accession of new members, the EU9 group remains constant over time, including those countries that had joined by 1973 (Belgium, Luxembourg, Denmark, France, Germany, the Netherlands, Ireland, Italy and the UK). It is obvious that EU exports have grown very rapidly over the period, with particular growth spurts during the second half of the 1970s and 1980s, as well as over a few years in the mid-1990s.

The short-term growth rates have obviously been affected by global macroeconomic developments, with events like the second oil crisis in the early 1980s and the ERM crisis in the early 1990s leaving clear marks on performance. However, the main driving forces are likely to be related to income growth and trade liberalization: trade has expanded as trade partners have grown richer and tariff barriers have fallen throughout the 40-year period covered by Figure 1. However, the tariff reductions have not been uniformly distributed across regions and goods. Trade liberalization has proceeded faster for the relations between developed countries than for North-South trade. In particular, Europe has been able to reduce its internal trade barriers faster than the barriers to trade with other regions. In terms of goods, the liberalization of trade in manufactures has been faster than the liberalization of trade in agricultural products. These distinctions have also affected EU trade. Figure 2 shows how the geographical pattern of EU exports has evolved from the early the 1960s to 2000 (Appendix 1 defines the different geographical groupings used in the subsequent analysis). It

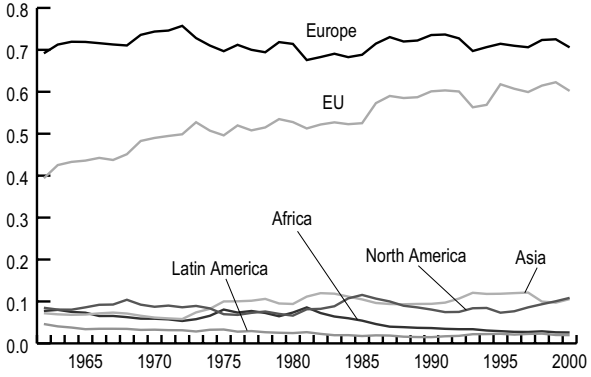
Figure 1 EU and EU9, total exports 1960-2000, current prices



Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris.

can be seen that intra-regional trade accounts for the bulk of EU's exports, and that it has increased greatly in importance during the period in question, from about 40 percent to around 60 percent of total exports. Two observations can be highlighted. Firstly, the period from 1962 to 1973 exhibits very rapid increases in the share of intra-EU trade, at the same time as the share of Europe as whole remains more or less constant. Since no new members were admitted during this period, there is reason to suspect some diversion of trade from non-members (mainly EFTA countries) to members. Secondly, the increase in the intra-EU share since the mid-1980s has coincided with the expansion of the European Union: trade with Europe as a whole, including members as well as non-members, has remained fairly constant at around 70 percent of total exports. North America has also maintained a fairly stable share of about 10 percent of exports, but the relative importance of Asia and Africa has changed notably. Asia increased its share from roughly 7 percent to over 10 percent (with a temporary dip after the onset of the Asian financial crisis in 1997), while the share of Africa declined from 7 percent to about 2.5 percent. Latin America has also seen a reduction in its share of EU exports. These changing shares largely reflect differences in the economic performance of the regions. Asian income growth has been high, and the Asian market has grown faster than other export markets. Africa and Latin America have not performed equally well, and their markets have become relatively less important over time.

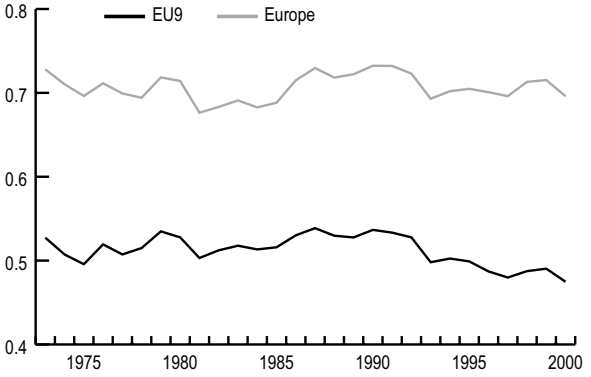
Figure 2 EU export shares according to destination



Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris.

Figure 3 looks closer at the development of intra-European trade since 1973, holding constant the group of exporting countries (focusing on the EU9). It can be seen that the share of intra-group exports for these countries has actually fallen more or less constantly since the mid-1980s, from a peak of about 54 percent to 47 percent in 2000, while the export share of

Figure 3 Export shares for EU9



Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris.

Europe as a whole has remained in the neighborhood of 70 percent. This confirms that the increasing share of intra-EU exports after 1973 in Figure 2 is essentially a result of the expansion of the EU.

Looking at the commodity structure of EU trade, it would be reasonable to expect the pattern of exports and imports to reflect the comparative advantages of the EU and its trade partners. Table 1 therefore shows how EU exports and imports were distributed across six broad industry groups – capital intensive (K), labour intensive (L), human capital intensive (H), technology intensive, land intensive (mainly agricultural) and energy intensive products – at four points in time between 1962 and 2000. Appendix 2 identifies the two-digit SITC Rev. 2 categories included in each industry group. The years are chosen to distinguish the three episodes of European integration discussed above. Apart from total exports and imports, Table 1 also distinguishes between intra-EU trade and trade with outsiders (non-EU trade).

Beginning with the pattern of exports, Table 1 confirms the importance of intra-regional trade noted in Figure 2: the share of intra-EU exports grew from less than 40 percent of total exports in 1962 to nearly 60 percent in 2000. The table also reveals some of the changes in the product structure of EU exports. Overall, capital and technology intensive products were dominant in 1962, with each of the product groups accounting for 27-28 percent of total exports. By 2000, technology intensive products had increased their share to 37 percent, while the share of capital intensive products had fallen to 16 percent. Simultaneously, human capital intensive products had emerged as the second largest group, with 22 percent of total exports in 2000. This pattern probably reflects the development of underlying production conditions, with continuous increases in the level of human capital and technological capability in the member states. In addition, the Northern expansion of the EU in 1995 added member countries (Austria, Finland, and Sweden) with substantial capacity in human capital intensive and technology intensive sectors. The implication, that the EU has comparative advantages in relatively advanced industries, is well in accordance with the stylized picture of the EU as a region with relatively abundant human capital and technological assets.

Looking separately at intra-EU and non-EU exports, some further observations emerge. Firstly, it is obvious that the increases in intra-EU exports have been concentrated to human capital and technology intensive products: these product groups have increased their shares of intra-EU exports as well as total EU exports. The intra-EU exports of capital intensive, land intensive, and energy intensive products have increased roughly in line

with the total increases in EU trade. However, the intra-EU exports of labour intensive products have grown twice as fast as EU's overall exports. This is interesting, because it is generally assumed that the EU does not have any comparative advantages in labour intensive production. A reasonable explanation is that the expansion of the EU during the 1980s brought in new member countries (Greece, Spain, and Portugal) with comparative advantages in labour intensive sectors. The development of non-EU exports corresponds well with the changes in intra-EU exports. The EU shows continuing strength in human capital and technology intensive products, but the shares of capital intensive, land intensive, and energy intensive products have fallen. Interestingly, non-EU exports of labour intensive products have almost maintained their share of total EU exports since 1962, which is also a likely consequence of the expansion of the EU to Greece, Spain, and Portugal during the 1980s.

Turning to the import side, a first point to note is that the shares of intra-EU and non-EU imports have remained relatively stable over time. In 1962, the intra-EU imports corresponded to approximately 52 percent of total EU imports: in 2000, the share was 54 percent. Capital intensive and land intensive products were the largest product groups in 2000, but technology intensive products were more important until the 1970s. The predominance of land intensive products is probably an indication of remaining trade barriers: there is no reason to believe that the EU would have substantial advantages in these product groups if the European market was completely open to foreign competition. However, it should be noted that the fluctuations in the shares of the different product groups are uncomfortably large, and it is likely that the quality of import data is significantly lower than that for export data. It is therefore difficult to make any precise statements regarding the changes in the product structure of intra-EU and non-EU imports.

Table 2 presents a more detailed picture of the regional orientation of EU's total exports in 1962 and 2000, distinguishing between geographical regions as well as industry groups. Considering the changes in EU membership, the table does not distinguish between EU members and other European countries: instead, Europe is treated as one group. The data on export shares refer to the export share for each specific industry group (hence, the first data entry in the table shows that 8 percent of all European exports of capital intensive products in 1962 were directed to Africa).

A closer look at Table 2 reveals a number of interesting observations. Firstly, for each region, the weights of the different industry groups are relatively evenly distributed. That is, the composition of export is relatively

Table 1 EU export and import shares per industry group (percent of total export and import)

	K int	L int	H Int	Tech int	Land int	Energy int	Total
Export 1962							
Intra-EU export	11.64	2.40	5.02	9.27	7.88	2.85	39.06
Non-EU export	15.84	3.80	10.75	19.09	7.39	3.01	59.88
Total export	27.48	6.20	15.77	28.36	15.27	5.86	98.94
Export 1973							
Intra-EU export	13.12	4.40	7.97	13.94	10.35	2.64	52.41
Non-EU export	10.96	3.06	9.50	16.78	4.86	1.47	46.63
Total export	24.08	7.46	17.47	30.71	15.21	4.10	99.04
Export 1986							
Intra-EU export	11.17	5.26	9.50	17.30	9.70	3.63	56.56
Non-EU export	7.79	4.13	7.76	16.39	4.12	1.66	41.85
Total export	18.95	9.39	17.26	33.70	13.82	5.29	98.41
Export 2000							
Intra-EU export	10.00	5.28	13.51	20.88	7.10	2.59	59.35
Non-EU export	6.04	3.20	8.82	16.63	2.91	1.42	39.02
Total export	16.04	8.48	22.33	37.51	10.01	4.01	98.37
Import 1962							
Intra-EU import	11.63	4.26	2.57	16.58	11.81	5.08	51.92
Non-EU import	5.45	6.79	1.56	9.35	8.93	10.33	42.41
Total import	17.08	11.05	4.13	25.93	20.74	15.41	94.33
Import 1973							
Intra-EU import	3.21	7.69	11.08	26.50	10.32	0.24	59.04
Non-EU import	1.21	4.37	9.22	15.17	9.88	0.87	40.72
Total import	4.41	12.06	20.31	41.67	20.20	1.11	99.76
Import 1986							
Intra-EU import	7.94	25.49	5.66	6.67	4.89	3.04	53.69
Non-EU import	4.23	11.27	5.65	3.41	5.27	11.17	40.99
Total import	12.17	36.76	11.31	10.08	10.16	14.20	94.68
Import 2000							
Intra-EU import	18.19	1.83	5.89	8.02	15.87	4.27	54.07
Non-EU import	7.11	1.09	8.25	5.19	11.04	9.99	42.67
Total import	25.30	2.92	14.14	13.21	26.91	14.26	96.74

Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris.
Note: The industry category "Other" (accounting for 1-2 percent of total exports and up to 6 percent of total imports) has been excluded from the Table. Total exports and imports therefore sum to less than 100 percent.

Table 2 Trade share per region of total export per industry

Region	Industry group	Export share 1962-63	Export share 1999-00	Change 1962-2000
Africa	K intensive	0.08	0.03	**-0.05**
	L intensive	0.09	0.02	**-0.07**
	H intensive	0.09	0.02	**-0.07**
	Tech intensive	0.07	0.03	**-0.04**
	Land intensive	0.08	0.03	**-0.05**
	Energy intensive	0.06	0.02	**-0.04**
Asia	K intensive	0.06	0.10	0.04
	L intensive	0.04	0.08	0.04
	H intensive	0.08	0.10	0.02
	Tech intensive	0.10	0.12	0.02
	Land intensive	0.05	0.09	0.04
	Energy intensive	0.01	0.03	0.02
Europe	K intensive	0.71	0.76	0.05
	L intensive	0.69	0.77	0.08
	H intensive	0.62	0.72	0.10
	Tech intensive	0.68	0.67	**-0.01**
	Land intensive	0.78	0.79	0.01
	Energy intensive	0.83	0.76	**-0.07**
North America	K intensive	0.10	0.08	**-0.02**
	L intensive	0.16	0.10	**-0.06**
	H intensive	0.12	0.11	**-0.01**
	Tech intensive	0.06	0.13	0.07
	Land intensive	0.06	0.05	**-0.01**
	Energy intensive	0.01	0.11	0.10
Oceania	K intensive	0.01	0.01	0.00
	L intensive	0.01	0.01	0.00
	H intensive	0.01	0.01	0.00
	Tech intensive	0.01	0.01	0.00
	Land intensive	0.00	0.00	0.00
	Energy intensive	0.00	0.00	0.00
Latin America	K intensive	0.04	0.01	**-0.03**
	L intensive	0.02	0.02	0.00
	H intensive	0.05	0.02	**-0.03**
	Tech intensive	0.07	0.02	**-0.05**
	Land intensive	0.02	0.01	**-0.01**
	Energy intensive	0.00	0.01	0.01

Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris.

similar across regions. The main exceptions are Asia and North America, where the share of technology intensive products is relatively large, and the shares of energy intensive products (Asia) and land intensive products (North America) are low. Secondly, the increasing importance of Asia and the diminishing role for Africa and Latin America is very clear. Asia has gained shares in all product groups, whereas Africa and Latin America have lost in almost every product group. Thirdly, Europe has become more important as a destination for most EU exports, with the notable exception of energy intensive products: there, North America has become a more important customer.[2]

3.2 EU trade with transition economies and developing countries

At the same time as Europe's internal trade barriers were reduced and eventually abolished within the framework of EU integration, trade with the rest of the world has also been liberalized. The most important developments have occurred at the multilateral level, where GATT and WTO have contributed to substantial reductions in formal tariffs: the average tariff for manufacturing goods in trade between OECD countries had fallen to 7 percent already by the late 1970s, after the Tokyo Round of GATT negotiations. In addition, the EU has entered into a large number of bilateral trade agreements with other regions and countries. The first preferential trade agreements were established in connection with the Treaty of Rome in 1957, and involved "countries and territories" that were associated to the member states: these were mainly former or current colonies in French-speaking Africa that were granted preferential access to the EU market, together with some development funds. The Yaoundé Conventions (1963 and 1969) extended these preferences to some 20 newly independent African states. After the accession of the UK, the agreement was expanded to cover Commonwealth countries in the ACP (Africa, Caribbean, and Pacific) region. This agreement, signed in 1975 and known as Lomé I, provided development aid, non-reciprocal preferences, and a system to stabilize fluctuating export earnings to 46 ACP countries. During the following 20 years, the Lomé convention was extended four times, with membership increasing to 70 ACP countries. The present agreement, covering 78 ACP countries, was signed in Cotonou in 2000. It is based on partnership agree-

[2] Energy intensive industries corresponds to SITC 32–35, Coal, coke and briquettes; Petroleum, petroleum products and related materials; Gas, natural and manufactured; Electric current.

ments that provide non-reciprocal free access to the EU market in most manufacturing sectors, as well as financial aid and technical support. In return, it requires the ACP countries to engage in a political dialogue with the EU on a number of sensitive issues, ranging from human rights to terrorism and non-proliferation issues. Another new component in the Cotonou agreement is explicit EU support to South-South regional integration.

Since 2001, the 50 least developed countries (LDCs) have an even more favorable agreement, the Everything But Arms (EBA) agreement. It promises duty-free access to the European market without any quantitative restrictions in all areas except arms and ammunition. Special transition rules are in place for sugar, rice, and bananas, where the quantitative restrictions must be removed before the end of 2009. Those developing countries that do not belong to either the ACP group or the LDC group still have preferential treatment through the generalized system of preferences (GSP), which provides more favorable treatment than what is guaranteed through WTO's Most Favored Nation (MFN) clause.

Other preferential trade agreements apply to the EU's nearest neighbors. The free trade agreement with EFTA, the EEA agreement, and the Association Agreements with the Central and Eastern European countries have already been mentioned. Although Austria, Finland and Sweden became EU members in 1995, and Switzerland decided not to join the EEA, it is still important for the three non-EU members of the agreement – Norway, Iceland and Liechtenstein. Moreover, the relations between the EU and Romania and Bulgaria will be based on Association Agreements until 2007, when these two countries accede to the Union. Other agreements cover relations with Turkey (customs union), the Western Balkans (stabilization and association agreements, with extensive free trade), Russia and Ukraine (partnership and cooperation agreements, including GSP preferences), Mediterranean countries (free-trade area to be completed 2010), and Chile and Mexico (free trade agreements), with negotiations underway to establish a free trade area also with the MERCOSUR. In fact, there are only a handful of countries that rely primarily on WTO's most favored nation rules – these are the US, Japan, Taiwan, Australia, Canada, and New Zealand (Panagariya 2002).

It can be argued that the prevalence of preferential trade agreements should reduce the concern for trade diversion from European integration, since almost all outsiders have been able to negotiate duty-free access to the European markets in most product categories. This means that reductions in internal tariffs will, per definition, not lead to lost trade opportunities for those efficient outside producers that have unrestricted access to

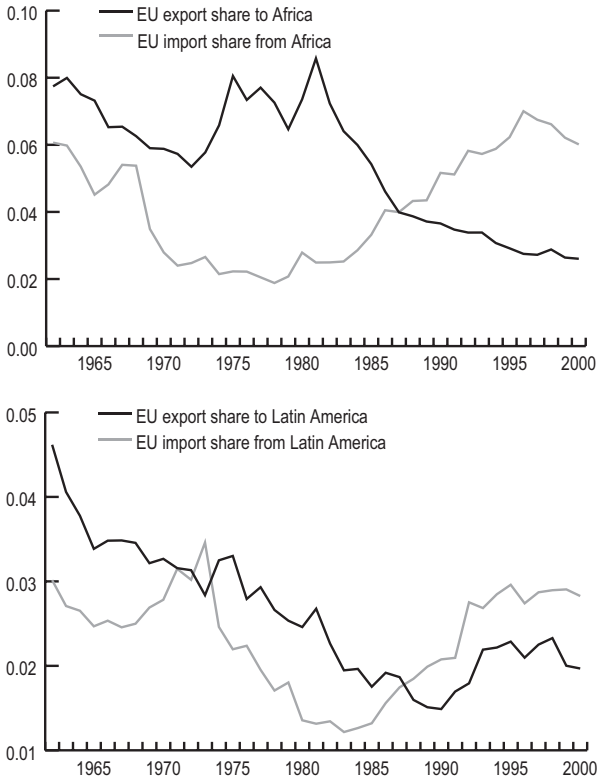
the internal market: the nominal tariff structure will not discriminate against outsiders.[3]

The same argument largely applies to those developed countries that have the least favorable bilateral treatment, relying only on MFN rules. The countries in this category are advanced industrial economies with the bulk of their exports in the manufacturing sector, and they have been among the greatest beneficiaries from multilateral trade liberalization: the average MFN tariffs on manufacturing are low, with EU tariffs for many product categories bound at zero.

However, at the same time as tariffs have lost much of their importance as instruments of protectionism in the manufacturing sector, they remain significant in EU's agricultural trade regime, where quantitative trade restrictions are also common. In addition, new kinds of non-tariff barriers to trade have emerged in the manufacturing sector during the past decades. National and regional product standards that are costly to satisfy for outsiders and government procurement practices favoring national or regional producers are only some examples of such measures. Rules of origin – specifying what share of a product's value added must originate in the exporting country in order to qualify for preferential treatment – may also limit the practical ability of outsiders to take advantage of seemingly generous trade preferences. Hence, the low nominal tariffs and the plethora of free trade agreements may give the illusion of a more liberal trade regime than what is actually the case. Furthermore, to the extent that various NTB are handled by the regional integration agreement but not by the bilateral or multilateral agreements regulating trade relations with the rest of the world, there is a risk that regional integration will distort international trade flows even if nominal external tariffs are very low. As noted above, the abolition of NTBs lies squarely at the center of the Single Market Program. NTBs have also been core issues in multilateral trade negotiations ever since GATT's Tokyo Round 1973-1979, but with significantly weaker results than in the SMP. It is therefore highly relevant to address the concerns about the effects of European integration on outsiders. The remainder of this section looks at the evolution of EU trade with developing regions, while the following sections turn to a closer examination of trade diversion.

[3] Outside producers may still lose market shares to firms from member countries if the integration process contributes to higher efficiency, e.g. as a result of tougher competition or stronger economies of scale. However, this does not constitute trade diversion in the sense of Viner (1950).

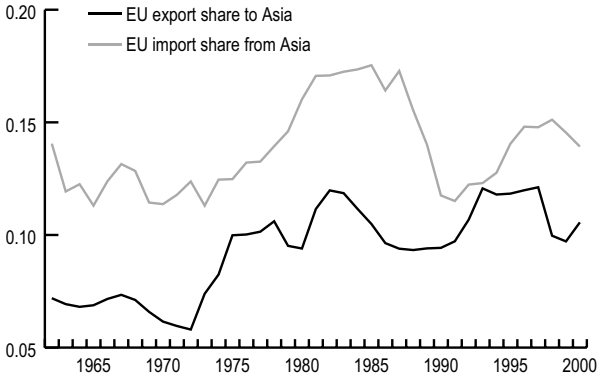
Figure 4 EU trade with Africa and Latin America 1962-2000



Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris.

Figure 4 illustrates the shares of Africa and Latin America in EU exports and imports between 1962 and 2000. It confirms the conclusion from Table 2, that both continents have become less important as destinations for EU exports, but also shows that imports from both regions have developed favorably, increasing from about four percent around 1980 to nine percent in 2000. It is likely that the preferences granted through the Lomé Conventions have contributed to this relatively good performance. The African share of EU exports was clearly above the diminishing long term trend during a ten-year period starting in the mid-1970s, which may

Figure 5 EU trade with Asia 1962-2000



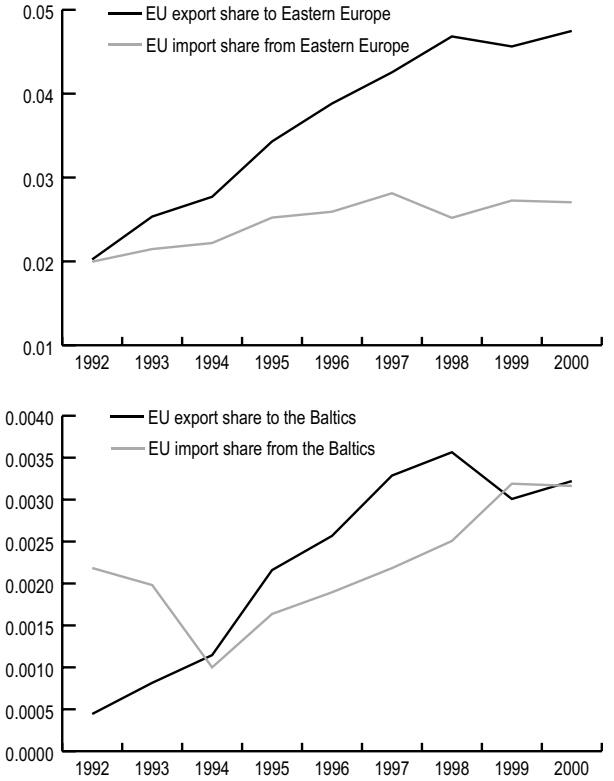
Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris.

reflect the UK accession to the EU, but any positive effect seems to have faded quite rapidly. The development of Latin American exports was equally weak until the late 1980s, but the 1990s saw somewhat faster export increases. This was probably due to the macroeconomic stabilization that occurred in large parts of the continent from about 1991, with Argentina and Brazil as the main examples of countries where inflation rates were brought down from several hundred percent per year to single digit levels. However, a fair conclusion is that both Africa and Latin America are of marginal importance for Europe's aggregate exports.

Figure 5 focuses on EU trade with Asia. The contrast with Africa and Latin America is obvious, both regarding Asia's export and import shares. The higher importance of Asia is obviously explained by the fact that its weight in overall world trade is higher than that of Africa and Latin America, at the same time as the economic performance of many Asian countries has been far better than that of other developing regions. The European trade deficit with Asia is concentrated to Japan and China, but the EU has recorded periodical trade surpluses with many of the other rapidly growing economies in East and Southeast Asia.

Figure 6 summarizes the data for trade relations with Eastern Europe and the Baltic states. These data series do not start until 1992, after the collapse of the Soviet Union, but it is clear from the figure how quickly trade relations have developed since then. Six of the Eastern European econo-

Figure 6 EU trade with Eastern Europe and the Baltics 1962-2000



Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris.

mies and all three Baltic states have been EU members since May 1, 2004, but it can be seen that the trade relations with the EU had already begun to develop in the early 1990s. In particular, EU exports have grown rapidly, both as a result of the underdevelopment of consumer goods industries in most of the planned economies and the investment needs stemming from the transition process. The growth of imports from Eastern Europe has been slower: it has taken time to develop the production capacity needed to compete successfully in EU markets. The Baltic states – in particular Estonia – are among the countries that have been most successful in redirecting

trade from East to West, as shown by the rapid increase in their share of EU imports.

Although Figures 4-6 only provide a very limited picture of how the trade relations between the EU and the developing countries have developed, they may still be interpreted as an indication that the recent stages of European integration have not had any dramatic negative consequences for developing countries. A caveat is that the share of imports from developing countries diminished during the first 15-20 years of EU integration, which could have been due to diversion of trade in favor of regional producers. However, it is clear that the trend has been reversed since the late 1970s and developing countries have managed to capture increasing shares of EU's import market, in spite of the deepening of EU integration during this period. While this does not prove that EU integration is harmless – the growth of developing country exports would perhaps have been even faster without EU integration – it suggests that it is necessary to examine the data much more closely in order to see how severe the problem of trade diversion is.

4 COMPARATIVE ADVANTAGE, REGIONAL ORIENTATION AND TRADE DIVERSION

The concept of trade diversion is intimately connected to patterns of comparative advantage. The situations where regional integration is potentially harmful for the global economy occur when efficient producers located outside the integrating region lose market shares to less efficient regional producers with privileged access to the regional market. This means that cases where efficient regional producers capture market shares from less efficient outsiders (e.g. developing country producers who had been able to capture parts of the market thanks to preferential trade rules) do not constitute harmful trade diversion. To distinguish between these cases and to examine whether there is trade diversion, it is necessary to define the comparative advantages of the integrating region. While the preceding tables give some impressions about the comparative advantages of the EU, they fail to account for the trade patterns of other countries. The simple fact that a large share of EU exports can be found in technology intensive products does not by itself mean that the EU has a comparative advantage in this area: Europe may import even more technology intensive products, or other countries may have even larger trade shares in high tech sectors. To identify comparative advantages, the first step is therefore to define a reference point against which to assess whether observed exports are relatively large or small.

The literature on international trade presents a number of alternative measures of comparative advantages. The "Revealed Comparative Advantage index" (RCA-index) has already been defined in Section 2 above in the discussion of MERCOSUR's trade performance. However, Hinloopen and Van Marrewijk (2001) observe that the mean of the distribution of RCA is well above unity, suggesting that each country or region has a comparative advantage in its average sector. This empirical result is the result of the chosen specification of the RCA index. Hoen and Oosterhaven (2004) suggest an alternative specification, where an additive RCA index is specified as

$$\textbf{A-RCA}_i = (x_{io}/X_o) - (x^*_{iw}/X^*_w) \qquad (3)$$

where (x_{io}/X_o) is the share of product group i in the country's or region's external trade and (x^*_{iw}/X^*_w) is the country's or region's share of total world exports, as before. The A-RCA index is symmetric around zero.

An alternative measure capturing comparative advantages is the Michelaye index (MI). The Michelaye index is defined as

$$\textbf{MI}_i = (x_{io})/X_o) - (m_{io}/M_o) \qquad (4)$$

where $(x_{io})/X_o$) is simply the export share of industry i in the country's or region's trade, and (m_{io}/M_o) is the import share. The values of the Michelaye index range from -1 to 1. A negative (positive) value in industry i indicates that a country or a region is under (over) specialized in that industry, while a value of zero indicates neutrality. Hence, while the A-RCA index uses the export structure of a reference group (for example, the rest of the world) as a point of comparison, the MI index is based on the country's own trade patterns. This means that the two indices may give different indications of relative advantages and competitiveness.

Table 3 tabulates the A-RCA index and MI index for the EU in the six broad industry groups discussed earlier at four points in time: 1962, 1974, 1986, and 2000. A major observation is that there are substantial differences between the two indices. For instance, in 2000, the EU had comparative advantages in capital intensive and land intensive sectors and comparative disadvantages in labour and energy intensive products according to both indices, but they disagreed about the status of the human capital intensive and technology intensive product groups. According to the MI-index, the EU had a mild disadvantage in human capital intensive products but a comparative advantage in technology intensive products: for both of these product groups, the year 2000 marks a break from earlier observations that indicated the opposite pattern. The A-RCA index, by contrast, suggests that the EU had comparative advantages in human capital intensive products, but that the past advantages in technology intensive products had disappeared by 2000. These differences are clearly related to both the import pattern of the EU and the trade patterns of other countries. The loss of comparative advantages in human capital intensive sectors suggested by the MI-index is mainly explained by increasing EU consumption and imports of human capital intensive products, while the loss of comparative advantages in technology intensive products, implied by the A-RCA-index, emanates from the great increase in Asian and North American production of these products. In this comparison, it can be argued that the A-RCA index gives a picture that is more consistent with the patterns given by Tables 1 and 2, so we will mainly use the A-RCA index in the subsequent discussion.[4]

[4] Another advantage is that the A-RCA index does not rely on import data, which tend to be less reliable.

Table 3 The Michelaye index and the A-RCA index for the EU, by industry

	MI-index	MI-index	MI-index	MI-index	Change
Industry group	**1962**	**1973**	**1986**	**2000**	**1962-2000**
H intensive	0.14	0.20	0.08	-0.01	-0.15
Tech intensive	-0.09	-0.04	-0.17	0.06	0.15
K intensive	0.14	-0.02	0.05	0.04	-0.10
Land intensive	0.11	-0.02	0.30	0.30	0.19
L intensive	-0.08	-0.14	-0.03	-0.17	-0.09
Energy intensive	-0.18	0.01	-0.22	-0.19	-0.01
	A-RCA index	A-RCA index	A-RCA index	A-RCA index	A-RCA index
Industry group	**1962**	**1973**	**1986**	**2000**	**1962-2000**
H intensive	1.00	0.60	0.26	0.21	-0.79
Tech intensive	0.49	0.30	0.17	-0.38	-0.87
K intensive	1.39	1.31	0.18	0.95	-0.44
Land intensive	1.04	1.26	1.17	0.62	-0.42
L intensive	-1.13	-0.83	-0.33	-0.07	1.06
Energy intensive	-1.82	-3.46	-3.61	-3.08	-1.26

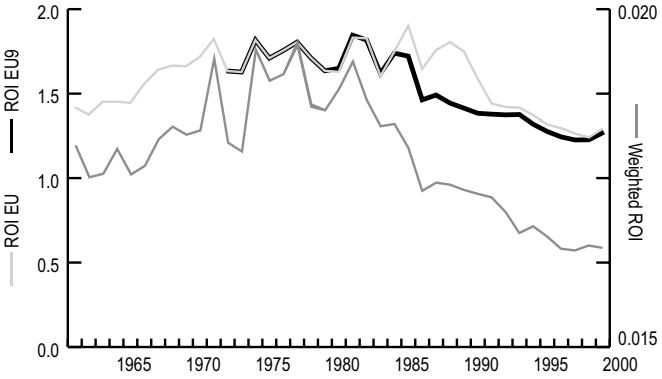
Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris, and Feenstra *et al*. (2005), *World Trade Flows: 1962-2000* (January 2005), NBER Working Paper No. W11040.

Notes: A positive (negative) value of the MI- and A-RCA-index indicates a comparative advantage (disadvantage). The A-RCA index is multiplied by 100 to facilitate comparison with the MI-index.

The simple correlation between the Michelaye index and the A-RCA index is positive (0.28) and significant (p-value = 0.00).

Apart from the comparative advantage measure, we also need a measure of the direction of trade: as proposed by Yeats (1998), we use the Regional Orientation Index (ROI), which measures how concentrated exports are to the regional market (see equation 1 above). Figure 7 provides a first look at how the regional orientation of EU exports has developed during the 40-year period under study. Just like in the MERCOSUR case, regional orientation increased substantially during the first decade of integration, but this development was halted after the mid-1970s. After the mid- to late 1980s, the regional orientation of trade has actually diminished, both for the EU as a whole and for the EU9 group. There are several possible explanations to the falling regional orientation during the past two decades. Firstly, the reduction in international trade costs coupled with global trade liberalization has made distant markets more accessible to European firms. Secondly, the growth rates of most EU economies have been well below the

Figure 7 Regional orientation of exports, EU and EU9 1962-2000



Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris.
Note: ROI calculated as average ROI across industries per year. Weighted ROI is weighted by industry share of EU intra and inter export.

growth rates of the US and a number of Asian countries in recent decades. Hence, the relative size of the European market has decreased, and the expansion of European industry has increasingly been directed to dynamic overseas markets, particularly in East Asia. Thirdly, it should be noted that the explicit objective of the later stages of the European integration process (since the mid-1980s) has been to strengthen the competitiveness of European firms by raising the level of competition and encouraging the development of larger firms that are able to exploit economies of scale (Gustavsson and Kokko 2004). If this objective is achieved, EU based firms should improve their ability to penetrate markets outside the EU. Hence, the somewhat paradoxical outcome may be that the more advanced stages of European integration have actually reduced the regional orientation of EU-based firms' exports.[5]

[5] As a robustness test, the Figure also includes a weighted ROI-index, where the weight of each industry is proportional to its share of total EU exports. This does not change the inverse U-shape of the ROI-index: instead, it tends to make the decrease in regional orientation even more pronounced.

Table 4 illustrates the changes in the regional orientation of EU trade for the six broad industry groups used earlier. The downward trend in regional orientation is not as clear as for aggregate EU exports, but the mid-1980s seem to constitute a turning point here as well. In fact, there are signs of an inverse U-shape in all industry groups except human capital intensive goods.

Table 4 Regional orientation of EU trade, per industry group

Industry group	ROI 1962	ROI 1973	ROI 1986	ROI 2000	Growth 1962-2000, percent	Growth 1986-2000, percent
K intensive	1.15	1.27	1.23	1.17	2	-5
L intensive	1.08	1.40	0.96	1.14	6	19
H intensive	0.68	0.72	0.78	0.87	27	11
Technology intensive	0.71	0.72	0.84	0.81	15	-3
Land intensive	2.12	2.34	3.18	1.70	-20	-47
Energy intensive	2.71	4.22	3.69	2.28	-16	-38

Source: OECD (2002), *International Trade by Commodity Statistics*, OECD, Paris

At a disaggregated level, neither A-RCA nor ROI provides much useful information when observed in isolation or at a single point in time. However, the development of comparative advantages and regional orientation over time may reveal interesting information about changes in trade patterns, as noted by Yeats (1998). Tables 5–7 provide a first comparison with Yeats' (1998) analysis, identifying the ten two-digit industries (out of 67) that exhibit the most pronounced increase in the ROI and regional trade at various times during the 1962–2000 period. The three time periods are 1962–1973, 1974–1986, and 1987–2000, which correspond to the three stages of EU integration discussed in section 3 above. Apart from this focus on different stages of regional integration, the main differences between our analysis and Yeats' original contribution are the level of aggregation and the definition of comparative advantages – Yeats focused on the 30 three-digit sectors with the largest increases in ROI while we select ten two-digit industries, and Yeats used RCA whereas we use A-RCA.

The industries included in part a of each table are selected on the basis of the relative increase in ROI during the period in question; this corresponds to the selection criteria used by Yeats (1998). Part b includes those industries that had the largest absolute increases in ROI. Part c, by contrast, focuses on the ten industries with the largest absolute increases in intra-EU trade (as in Berkovica *et al*. 2003). The reasons for using several alterna-

Table 5 Changes in intra-regional trade and A-RCA 1962-1973

a) TOP 10 industries: fastest growth in ROI 1962-1973 (groi6273)

Industry	Class	A-RCA	d.A-RCA	ROI	d.ROI	g.ROI	ROIM	d.ROIM	EU-X	d.EU-X
Gas	Energy int	-0.39	-0.30	9.93	7.15	256.76	0.08	-0.01	0.52	558 064
Oils-fats	Land int	0.04	0.00	1.62	1.13	232.09	1.40	0.54	0.15	154 044
Animal oils	Land int	-0.16	0.02	4.89	3.15	181.15	0.27	0.12	0.08	76 918
Meat	Land int	-1.08	-0.04	4.22	2.67	171.44	0.97	-2.65	3.06	3 198 444
Fertilizers	Tech int	-0.03	-1.24	0.80	0.49	153.32	1.12	-0.04	0.42	395 922
Live animals	Land int	-0.33	0.28	8.32	4.45	115.13	0.91	0.69	1.11	1 182 762
Vegetable oils	Land int	-0.41	0.15	1.58	0.81	105.99	0.13	-2.29	0.32	330 261
Cork and wood	K int	-0.59	-0.47	2.28	1.13	98.50	0.59	-0.36	0.60	600 941
Petroleum etc	Energy int	-12.83	-5.81	1.23	0.61	97.55	1.21	-0.73	3.39	3 401 270
Oil seeds	Land int	-1.33	-0.03	2.75	1.32	91.74	0.55	0.21	0.10	103 625
								Sum	**9.77**	

b) TOP 10 industries; largest absolute increase in ROI 1962-1973 (droi6373)

Industry	Class	A-RCA	d.A-RCA	ROI	d.ROI	g.ROI	ROIM	d.ROIM	EU-X	d.EU-X
Gas	Energy int	-0.39	-0.30	9.93	7.15	256.76	0.08	-0.01	0.52	558 064
Live animals	Land int	-0.33	0.28	8.32	4.45	115.13	0.91	0.69	1.11	1 182 762
Animal oils	Land int	-0.16	0.02	4.89	3.15	181.15	0.27	0.12	0.08	76 918
Meat	Land int	-1.08	-0.04	4.22	2.67	171.44	0.97	-2.65	3.06	3 198 444
Oil seeds	Land int	-1.33	-0.03	2.75	1.32	91.74	0.55	0.21	0.10	103 625
Oils-fats	Land int	0.04	0.00	1.62	1.13	232.09	1.40	0.54	0.15	154 044
Cork and wood	K int	-0.59	-0.47	2.28	1.13	98.50	0.59	-0.36	0.60	600 941
Vegetables and fruit	Land int	-1.82	-0.67	3.30	1.06	47.30	0.40	-6.73	2.22	1 859 506
Vegetable oils	Land int	-0.41	0.15	1.58	0.81	105.99	0.13	-2.29	0.32	330 261
Clothing accessories	L int	-0.37	-1.36	1.94	0.79	68.35	2.67	-0.23	3.03	3 044 601
								Sum	**11.20**	

c) TOP 10 industries: largest absolute increase in intra-EU exports 1962-1973 (deu6273)

Industry	Class	A-RCA	d.A-RCA	ROI	d.ROI	g.ROI	ROIM	d.ROIM	EU-X	d.EU-X
Road vehicles	H int	4.97	-0.95	0.86	0.07	8.37	0.63	-0.25	9.39	9 380 508
Iron and steel	K int	3.20	-2.79	0.95	-0.25	-20.73	2.06	-2.05	6.52	5 804 723
Textile etc	K int	1.20	-1.12	1.25	0.02	1.25	0.94	0.16	6.01	5 562 463
Electrical mach nes	Tech int	1.47	-0.83	1.07	0.33	44.80	2.16	0.52	4.22	4 259 135
Industrial machinery	Tech int	3.35	0.52	0.72	-0.10	-11.92	0.75	-0.24	4.11	3 998 967
Specialized machinery	Tech int	4.91	2.70	0.50	-0.28	-35.62	0.76	-0.04	3.90	3 662 095
Petroleum etc	Energy int	-12.83	-5.81	1.23	0.61	97.55	1.21	-0.73	3.39	3 401 270
Meat	Land int	-1.08	-0.04	4.22	2.67	171.44	0.97	-2.65	3.06	3 198 444
Clothing accessories	L int	-0.37	-1.36	1.94	0.79	68.35	2.67	-0.23	3.03	3 044 601
Non-metallic mineral	K int	1.86	0.55	0.80	-0.15	-15.36	0.61	-0.39	3.03	3 000 687
								Sum	**46.65**	

Notes: d.A-RCA: absolute change in value of A-RCA index
d.ROI: absolute change in value of ROI index
g.ROI: relative change in ROI index (percent)
ROIM: regional orientation of imports
d.ROIM: absolute change in ROIM index
EU-X: share of EU exports
d.EU-X: absolute change in EU exports, current USD million

tive measures of regional orientation are, firstly, that industries with large relative increases in ROI may still have low regional orientation, and secondly, that the product groups with the largest (relative or absolute) increases in ROI may be relatively unimportant in overall trade. In addition to information about A-RCA, ROI, and changes in intra-EU trade, the tables also include data on the regional orientation of imports and the share of each industry in intra-EU trade.

Parts a and b of Table 5 (focusing on the period 1962–1973) provide a picture that seems to correspond relatively well with the concerns discussed by Yeats (1998). All but one of the industries with the largest increases in ROI had negative A-RCA values in 1973; most of these also experienced diminishing external competitiveness during the period. All but one of the industries were also oriented towards the EU market in 1973 (i.e., their ROI values exceeded 1). Looking more closely at the products included in parts a and b of Table 5, most are agricultural goods or raw materials – products where the trade barriers were higher than average, and where it is unlikely that the EU had any strong comparative advantages that were not related to trade regulations or other man-made obstacles to foreign competition. These observations suggest that the early stages of European integration may have contributed to some trade diversion, in particular through the common agricultural policy that protected the regional market from outside competition.

The columns labelled ROIM and dROIM provide some further information about the origin of imports. In most of the product groups, the regional orientation of imports was below unity in 1973, indicating that most imports were sourced from outside the EU. Moreover, the ROIM diminished in more than half of the product groups between 1962 and 1973. This confirms that the strongest effects occurred in areas where the EU had relatively weak competitiveness, and that integration probably helped local firms expand in industries where they would otherwise have been unable to compete.

One problem connected to the interpretation of the results discussed above is that the industries included in parts a and b of Table 5 only cover about 10 percent of intra-EU exports. This raises doubts about the possibilities to generalize the findings to a more representative set of industries. Part c of the table therefore examines the 10 industries with the largest absolute increases in intra-EU exports. This sub-sample covers nearly half of all intra-EU export in 1973, and provides a somewhat less worrying picture than the first two parts of the table. A first point to note is that most of the industries with large increases in the volume of intra-EU trade exhibited sub-

stantial comparative advantages in 1973, and were found in sectors exploiting EU's relatively abundant supplies of human and physical capital and technology. The pattern regarding the regional orientation of exports and imports was mixed, but in most sectors, the ROIM fell in spite of the increase in intra-regional exports. This indicates that the largest trade increases occurred in sectors with rapidly expanding consumption, allowing for growth in both intra-regional exports and shipments from other parts of the world.

Table 6 covers the period 1974-1986. Here, the picture is more unclear, both regarding the level of A-RCA in 1986, the changes in A-RCA over time, and the characteristics on the import side. One of the few possible generalizations is that fewer land intensive products appeared among the industries with large increases in regional orientation during this period; instead, regional trade seemed to gravitate towards technology and human capital intensive products like chemicals, office machinery, and road vehicles. In contrast to the pattern during the first period, the EU recorded positive A-RCA values in many of these product groups, weakening the arguments related to trade diversion. However, intra-regional trade increased substantially in some product groups where the long-term comparative advantages of the EU turned out to be relatively weak, e.g. clothing accessories and textiles. One reason for the strong regional trade performance in these product groups was probably the accession of Spain and Portugal to the EU: both countries had strong comparative advantages in clothing and textiles during the 1980s.

The picture for the period 1987-2000 is similar, with weaker overall signs of trade diversion but with some sectors that may give cause for worry. Most of the industries appearing in Table 7 record positive revealed comparative advantages, and there are only a few cases where regional trade increases rapidly at the same time as A-RCA is negative and diminishing. The main exception is office machines, which recorded large absolute increases in intra-regional trade but substantial reductions in A-RCA. Another observation is that most of the industries appearing in Table 7 are much more advanced than those found in Table 5, with scale economies related to capital requirements and fixed costs for R&D as the main determinants of competitiveness, and very low nominal protection levels. One of the main objectives of the SMP has been to raise the competitiveness in these sectors, and the findings in Table 7 suggest that the results may have been positive: although few of the industries in Table 7 record increasing external competitiveness, most have A-RCA values above zero and they have apparently been able to raise their regional sales without the assistan-

Table 6 Changes in intra-regional trade and A-RCA 1974-1986

a) TOP 10 industries: fastest growth in ROI 1974-1986 percent (groi7486)

Industry	Class	A-RCA	d.A-RCA	ROI	d.ROI	g.ROI	ROIM	d.ROIM	EU-X	d.EU-X
Oil seeds	Land int	-0.38	0.71	33.56	32.29	2 553.94	0.98	0.42	0.29	1 194 792
Explosives	Tech int	-0.01	-0.06	0.79	0.56	251.22	3.43	0.36	0.04	154 066
Fertilizers	Tech int	-0.39	-0.68	1.58	0.96	155.81	1.56	0.51	0.40	1 006 122
Arms and ammunition	Tech int	0.31	0.28	0.19	0.09	86.02	0.64	-0.59	0.07	260 734
Road vehicles	H int	0.94	-4.56	1.11	0.42	59.95	0.89	0.27	11.16	40 400 000
Animals	Land int	0.00	0.01	1.40	0.39	38.69	1.21	-0.09	0.01	16 152
Coal	Energy int	-1.05	-0.41	3.72	0.95	34.32	1.27	0.40	0.44	299 598
Plastic mtrl etc	Tech int	1.43	-0.77	1.39	0.27	24.51	0.99	-0.17	3.43	10 800 000
Rubber nes.	K int	0.32	-0.29	1.22	0.23	23.01	2.36	0.54	1.08	3 435 373
Crude rubber	Land int	-0.26	0.15	1.52	0.26	20.79	0.16	-0.10	0.30	896 439
								Sum	**17.20**	

b) TOP 10 industries; largest absolute increase in ROI 1974-1986 (droi7486)

Industry	Class	A-RCA	d.A-RCA	ROI	d.ROI	g.ROI	ROIM	d.ROIM	EU-X	d.EU-X
Oil seeds	Land int	-0.38	0.71	33.56	32.29	2 553.94	0.98	0.42	0.29	1 194 792
Fertilizers	Tech int	-0.39	-0.68	1.58	0.96	155.81	1.56	0.51	0.40	1 006 122
Coal	Energy int	-1.05	-0.41	3.72	0.95	34.32	1.27	0.40	0.44	299 598
Explosives	Tech int	-0.01	-0.06	0.79	0.56	251.22	3.43	0.36	0.04	154 066
Road vehicles	H int	0.94	-4.56	1.11	0.42	59.95	0.89	0.27	11.16	40 400 000
Animals	Land int	0.00	0.01	1.40	0.39	38.69	1.21	-0.09	0.01	16 152
Plastic mtrl etc	Tech int	1.43	-0.77	1.39	0.27	24.51	0.99	-0.17	3.43	10 800 000
Crude rubber	Land int	-0.26	0.15	1.52	0.26	20.79	0.16	-0.10	0.30	896 439
Office machines	Tech int	-0.25	-0.54	1.56	0.24	17.83	1.16	-0.53	3.81	14 900 000
Rubber nes	K int	0.32	-0.29	1.22	0.23	23.01	2.36	0.54	1.08	3 435 373
								Sum	**20.95**	

c) TOP 10 industries: largest absolute increase in intra-EU exports 1974-1986 (deu7486)

Industry	Class	A-RCA	d.A-RCA	ROI	d.ROI	g.ROI	ROIM	d.ROIM	EU-X	d.EU-X
Road vehicles	H int	0.94	-4.56	1.11	0.42	59.95	0.89	0.27	11.16	40 400 000
Electrical mach nes	Tech int	0.46	-1.27	0.93	-0.14	-13.36	0.67	-1.45	4.72	15 400 000
Office machines	Tech int	-0.25	-0.54	1.56	0.24	17.83	1.16	-0.53	3.81	14 900 000
Petroleum etc.	Energy int	-11.05	13.35	1.32	0.10	7.79	0.30	-0.52	4.60	14 200 000
Textile etc.	K int	0.12	-1.36	1.32	0.02	1.55	0.53	0.08	4.25	11 600 000
Industrial machinery	Tech int	3.74	0.02	0.62	-0.08	-10.87	1.21	0.46	3.77	11 500 000
Misc manuf nes	L int	1.58	0.22	0.75	-0.24	-23.84	1.25	-1.28	3.37	11 300 000
Plastic mtrl, etc	Tech int	1.43	-0.77	1.39	0.27	24.51	0.99	-0.17	3.43	10 800 000
Clothing accessories	L int	-1.37	-1.04	1.29	-0.78	-37.72	3.56	1.14	3.10	10 100 000
Organic chemicals	Tech int	1.67	-0.24	1.08	-0.03	-2.43	2.38	1.39	3.20	9 536 726
								Sum	**45.41**	

Notes: d.A-RCA: absolute change in value of A-RCA index
d.ROI: absolute change in value of ROI index
g.ROI: relative change in ROI index (percent)
ROIM: regional orientation of imports
d.ROIM: absolute change in ROIM index
EU-X: share of EU exports
d.EU-X: absolute change in EU exports, current USD million

Table 7 Changes in intra-regional trade and A-RCA 1987-2000

a) TOP 10 industries: fastest growth in ROI 1987-2000 percent (groi8700)

Industry	Class	A-RCA	d.A-RCA	ROI	d.ROI	g.ROI	ROIM	d.ROIM	EU-X	d.EU-X
Arms and ammunition	Tech int	0.13	-0.58	0.16	0.14	842.16	1.27	0.77	0.02	247 362
Coin	NEC	-0.01	-0.05	0.27	0.17	171.23	0.64	0.27	0.00	-8 097
Iron and steel	K int	-0.19	-1.32	1.53	0.61	66.84	1.63	0.24	3.25	22 000 000
Fish	Land int	-0.86	0.39	2.97	1.05	54.38	0.56	0.05	0.62	4 194 876
Metalworking mach	Tech int	0.27	-0.44	0.67	0.21	46.62	0.73	-0.11	0.62	4 286 338
Sugar etc	Land int	0.01	-0.01	0.94	0.29	45.40	0.87	0.28	0.23	1 497 959
Oils-fats	Land int	0.00	0.00	1.60	0.37	30.23	1.24	0.95	0.07	420 217
Pharmaceutical products	Tech int	2.89	1.51	0.68	0.16	29.34	2.04	1.03	2.61	28 700 000
Rubber nes	K int	0.07	-0.26	1.55	0.33	27.46	0.80	-1.75	1.06	7 927 844
Road vehicles	H int	1.04	-0.63	1.45	0.30	25.61	0.89	0.07	13.42	111 000 000
								Sum	**21.89**	

b) TOP 10 industries; largest absolute increase in ROI 1987-2000 (droi8700)

Industry	Class	A-RCA	d.A-RCA	ROI	d.ROI	g.ROI	ROIM	d.ROIM	EU-X	d.EU-X
Fish	Land int	-0.86	0.39	2.97	1.05	54.38	0.56	0.05	0.62	4 194 876
Iron and steel	K int	-0.19	-1.32	1.53	0.61	66.84	1.63	0.24	3.25	22 000 000
Coal	Energy int	-0.48	0.40	4.98	0.55	12.36	1.78	0.46	0.08	-768 844
Oils-fats	Land int	0.00	0.00	1.60	0.37	30.23	1.24	0.95	0.07	420 217
Rubber nes	K int	0.07	-0.26	1.55	0.33	27.46	0.80	-1.75	1.06	7 927 844
Road vehicles	H int	1.04	-0.63	1.45	0.30	25.61	0.89	0.07	13.42	111 000 000
Sugar etc.	Land int	0.01	-0.01	0.94	0.29	45.40	0.87	0.28	0.23	1 497 959
Coffee, tea, cocoa	Land int	-0.22	0.90	1.72	0.25	17.19	0.28	0.11	0.50	3 229 405
Footwear	L int	-0.26	-0.25	1.37	0.24	20.77	2.10	-1.18	0.77	4 403 918
Manuf metal nes	K int	0.57	-0.60	1.14	0.23	25.00	4.44	2.46	2.66	20 800 000
								Sum	**22.64**	

c) TOP 10 industries: largest absolute increase in intra-EU exports 1987-2000 (deu8700)

Industry	Class	A-RCA	d.A-RCA	ROI	d.ROI	g.ROI	ROIM	d.ROIM	EU-X	d.EU-X
Road vehicles	H int	1.04	-0.63	1.45	0.30	25.61	0.89	0.07	13.42	111 000 000
Electrical mach nes	Tech int	-2.30	-2.34	0.77	-0.17	-18.43	0.86	0.07	6.49	59 400 000
Office machines	Tech int	-3.01	-2.41	1.63	0.00	-0.02	1.06	-0.10	5.51	50 700 000
Telecommunications	H int	0.00	2.17	0.94	0.13	15.53	1.63	0.61	4.47	50 300 000
Industrial machinery	Tech int	2.61	-0.77	0.83	0.16	23.64	1.60	0.29	4.57	39 000 000
Misc manuf nes	L int	0.45	-0.91	0.90	0.12	16.06	1.02	-0.22	3.72	29 400 000
Pharmaceuticals	Tech int	2.89	1.51	0.68	0.16	29.34	2.04	1.03	2.61	28 700 000
Paper etc	K int	0.66	1.67	1.54	-0.35	-18.31	1.74	-0.58	2.92	27 000 000
Petroleum etc	Energy int	-10.38	1.34	1.13	-0.04	-3.03	0.10	-0.27	3.47	25 000 000
Plastic mtrl etc	Tech int	0.46	-0.96	1.53	0.13	9.35	1.03	-0.04	3.25	23 100 000
								Sum	**50.42**	

Notes: d.A-RCA: absolute change in value of A-RCA index
d.ROI: absolute change in value of ROI index
g.ROI: relative change in ROI index (percent)
ROIM: regional orientation of imports
d.ROIM: absolute change in ROIM index
EU-X: share of EU exports
d.EU-X: absolute change in EU exports, current USD million

ce of substantial trade barriers. In all, the general impressions from Table 7 do not support the results found by Yeats (1988) for MERCOSUR trade.

Looking at Tables 5-7 together, there are some indications that the effects of regional integration on trade may differ depending on the character of the integration agreement and the stage of regional integration at the outset. The results of the early stages of European integration, shown in Table 5, are fairly similar to those found by Yeats (1988), with large relative increases in intra-regional trade in sectors with weak external comparative advantages. However, at later stages of integration, the intra-regional export growth has increasingly been concentrated to areas with external comparative advantages. Moreover, looking at absolute rather than relative increases in intra-regional exports, there are no strong signs of trade diversion even at the early stages of EU integration: incidentally, this is the same results as that found by Berkovica *et al*. (2003) for absolute increases in intra-MERCOSUR exports.

4.1 What is the relation between ROI and A-RCA?

One drawback of Yeats' (1998) analysis, in light of the evidence for the later stages of European integration, is that it assumes that there is no underlying relation between changes in ROI and changes in A-RCA. Instead, the basic assumption is that regional integration will typically lead to higher regional orientation in trade, and the tabulations aim at detecting those cases where increasing ROI and decreasing A-RCA coincide. These cases are interpreted as potential examples of trade diversion: how else could industries with diminishing external competitiveness expand their regional trade. However, advanced integration agreements, like the SMP, are based on other assumptions about the effects of integration. In particular, the SMP assumes that integration will allow regional firms to exploit economies of scale and improve their competitiveness, which may cause either increases or reductions in regional orientation, depending on the initial level of external competitiveness. In industries with an initial comparative disadvantage (A-RCA < 0), it is possible that an increase in competitiveness due to regional integration will allow insider firms to substitute for imports from outside the region, thus raising the regional orientation of trade. It should be noted that this may occur without any trade diversion effects, if the reason for the increasing market shares is higher efficiency rather than explicit trade preferences. However, for industries with initial comparative advantages (A-RCA > 0), the pattern may be the opposite. Further increases in competitiveness may be associated with a decreased regional orientation, as the new market potential is primarily found outside of the integration area.

Table 8 Regression results. ROI and the A-RCA index. Dependent variable: ROI index, 1963-2000, 67 industries

Variable (t-value)	OLS 1962-2000	GLS AR1-error 1962-2000	GLS AR1-error 1962-1973	GLS AR1-error 1974-1986	GLS AR1-error 1987-2000
A-RCA $_{t-1}$ < 0	9.600 (3.86) [4.37]	9.709 (3.98)	0.414 (0.07)	14.707 (3.97)	12.617 (2.31)
A-RCA $_{t-1}$ > 0	-15.777 (-3.30) [-10.48]	-16.012 (-3.37)	-5.056 (-0.73)	-16.536 (-1.90)	-27.033 (-2.88)
K int Dummy	-2.301 (-11.08) [-7.90]	-2.309 (-11.29)	-1.688 (-5.41)	-3.767 (-9.22)	-1.687 (-5.39)
L int Dummy	-2.380 (-10.84) [-8.08]	-2.387 (-11.05)	-1.557 (-4.71)	-3.867 (-8.90)	-1.843 (-5.62)
H int Dummy	-2.727 (-12.16) [-9.33]	-2.730 (-12.36)	-2.245 (-6.40)	-4.212 (-9.27)	-1.927 (-5.77)
Tech int Dummy	-2.731 (-13.49) [-9.32]	-2.738 (-13.73)	-2.236 (-7.31)	-4.205 (-10.48)	-1.946 (-6.36)
Land int Dummy	-1.226 (-6.64) [-4.06]	-1.241 (-6.83)	-0.612 (-2.30)	-2.547 (-7.06)	-0.800 (-2.79)
Period dummies	Yes	Yes	Yes	Yes	
R^2	0.17				
No. of obs.	2 470	2 470	715	845	910
p-value	0.000	0.000	0.000	0.000	0.000

Notes: t-values in parentheses ( ). White corrected standard errors in square brackets [ ].
GLS allows for serial correlated and heteroscedastic standard errors.
Energy-intensive industries are used as reference for industry dummies.

In Table 8, we test this hypothesis by regressing the degree of regional orientation at time t on the A-RCA index at time t-1. The results support the hypothesis of an inverse U-shaped relation between regional orientation and the A-RCA index. The coefficient of initial A-RCA < 0 is positive and significant, while that of initial A-RCA > 0 is negative and significant: weakly competitive industries tend to increase their ROI over time, while

sectors that already have strong competitiveness reduce their regional orientation. This relation appears to become stronger over time: in fact, for the first integration period 1962-1973, the coefficients are not significant, but become increasingly larger and more significant during the second and third period.

The implication of these findings is probably that the risk for trade diversion à la Yeats is likely to be higher for industries with moderate A-RCA levels. The initially most competitive industries may have satisfied much of the regional demand, so that it is difficult to expand intra-regional trade any further. The least competitive industries, on the other hand, are unlikely to focus on external customers as their competitiveness increases. The industries with A-RCA levels somewhere between these two extremes are therefore the ones that are likely to be most interesting for the analysis. However, it should be noted that this discussion has highlighted some questions regarding the definition of trade diversion. Trade diversion has been used in the present analysis, as well as in earlier studies, to distinguish cases where regional producers capture market shares from more efficient outsiders thanks to regional trade preferences. This is essentially a static concept, since the relative efficiency of producers inside and outside the RIA is assumed to remain constant. Yet, modern regional integration (e.g. the SMP) is essentially a dynamic process, where the main effects of integration are related to growth rather than static resource allocation. It will be a challenge for future work to define a dynamic counterpart to the static concept of trade diversion.

4.2 How do these results compare with other studies?

The result that the early stages of European integration may have led to more trade diversion than the period after the establishment of the Single Market Program is not self-evident, and it is relevant to relate it to some of the other studies of European integration. One of the earliest ex post analyses is provided by Aitken (1973), who uses a gravity model to analyze bilateral trade flows for both the EC and EFTA. He accounts for transportation costs, income levels between trading partners, and institutional factors, such as the formation of trading blocks or a customs union and shows that both the EC and EFTA resulted in gross trade creation, with the effect being much larger for the EC than for EFTA. However, both the EC and EFTA resulted in mutual trade diversion during 1960s. Similarly, Bayoumi and Eichengreen (1995) report that the EC and the EFTA were trade creating among its member countries, and in the former case very rapidly so. They further report that the trade between the EC and EFTA fell by a

cumulated 17 percent between 1959 and 1964. They attribute this decline to the formation of the EC, as the EFTA became only fully operational in 1965 (although it was already established in 1960). In addition, they estimate that trade between the EC and the rest of the world was 1.7 percent per annum lower than expected from their gravity model, which suggests that the EC was trade diverting with respect to the rest of the world.

Similarly, Balassa (1975) found both trade creation and trade diversion, but a positive net effect of the two. He estimated the gross welfare gains for the European Community at about USD 0.7 billion per annum (equivalent of 0.15 percent of the EC GDP). The economic costs of trade diversion (largely due to the Common Agricultural Policy) were estimated to be in the range of USD 0.3-0.4 billion per annum, leading to a net welfare gain of about 0.4 billion per annum. Hence, the results from our analysis of development in ROI and A-RCA seem consistent with the finding that trade diversion was significant in the early years.

In contrast to the early years, the first enlargement seems to have been more trade creating. Prior to the accession of Denmark, Ireland and the United Kingdom, trade between the EC and these three countries was falling. After the accession, trade among the former three EFTA countries declined, due to a reorientation towards the original six EC member countries. Bayoumi and Eichengreen (1995) attribute this to the fact that joining the EC did not bring about any direct positive benefits for trade among the former three FTA countries. According to their calculations, between 60 and 90 percent of the subsequent trade increase between Denmark, Ireland and the United Kingdom and the EC-6 was due to trade creation. The second enlargement was strongly trade creating, aided by simultaneous economic liberalization in the new member countries, Greece, Portugal and Spain. According to Bayoumi and Eichengreen (1995), two-thirds to three-quarters of the trade increase is estimated to be due to trade creation in the case of Greece. For Portugal and Spain, a large share of the resulting trade increase is due to the increased trade between the two countries. Studies looking explicitly at trade diversion during this period are scarce.

The assessments of the economic effects of the Single Market Program have focused more on growth and investment effects than only on trade effects. The expectations before the completion of the SMP were high. The Emerson *et al*., (1988) report to the European Commission suggested that the increase in trade due to trade cost reduction would be remarkable. It was estimated that intra-EU trade would increase by 22.1 percent if the main effect of integration was a substantial reduction in trade costs, which in turn would create stiffer competition, resulting in lower prices and

higher output per firm. In this environment, the expected welfare gains were closely correlated with the degree of returns to scale, and the efficiency increases resulting from an increase in producer concentration and average firm size. However, the trade implications of full integration of the EU market were distinctly different. This difference stems from the fact that firms normally have larger market shares and some market power in their domestic markets, which motivates higher prices in the domestic market than in foreign markets. Two main effects were expected to arise if the "home-market effect" disappeared. Firstly, intra-EC trade would be reduced, as consumers would switch from imported varieties to price-reduced domestic varieties. Secondly, there would be a large positive welfare effect caused by price reductions. The welfare effects were expected to be approximately three times larger than in the comparable scenario, where market segmentation remains, both in the long and the short run. The total welfare effects were estimated to vary between industries, in the range of 0.5-2.0 percent in the case where trade costs are reduced, and in the range of 0.5-4 percent in the case of full market integration (Emerson *et al*. 1988). Several studies have also pointed out that the SMP is likely to influence investment flows and industry location, but the trade effects related to location models are often ambiguous (Baldwin *et al.* 1994, Forslid *et al*. 2002). The analyses typically predict concentration of production to the most favored locations – often the largest market – which tends to reduce trade in final goods, but may stimulate trade in intermediate products: the net effect of these two opposite tendencies may be positive or negative.

Ex post evaluations of the SMP argue that the integration effects are visible. The European Commission estimated in the mid-1990s that EU GDP was 1.1-1.5 percent above the level that would have prevailed without the SMP. Looking more specifically at trade effects, Krueger (1999) suggests that the EU boosted both intra-regional trade and trade between EU members and outsiders 1987-1997. Similarly, Cernat (2001) concludes that the EU had a very strong positive impact on trade between member states as well as trade with outsiders during the period 1994–1998. Hence, these studies show no signs of trade diversion, which may well be related to the substantial growth effects of the SMP. The trade diversion that probably occurred may simply have been overshadowed by increases in import demand related to growth.

Looking at longer time periods, Kandogan (2005) and Adams *et al*. (2003) provide somewhat contrasting pictures of the effects of EU integration. While Kandogan concludes that most of the EU's trade agreements have been trade creating both for the EU and for its partner countries in a varie-

ty of trade agreements, Adams *et al*. (2003) argue that the net effect of the EU has probably been trade diverting. The two studies apply distinctly different methodologies (although both depart from some variant of the gravity model), but the wide gap in conclusions illustrates that much remains to be done to reach a full consensus about the trade effects of regional integration.[6] This notwithstanding, it seems reasonable to conclude that different RIAs will have different trade effects and that different stages of any specific RIA are likely to generate different effects.

[6] It should be noted that Adams *et al*. (2003) observe that deeper integration, including liberalization of trade and investment rules, is likely to generate stronger positive effects on trade than simple agreements.

5 CONCLUSIONS

In this paper, we have applied the methodology of Yeats (1998) to examine the trade effects of European integration, with particular focus on the incidence of trade diversion. To this end, we have identified those industries that experienced the largest increases in the regional orientation of exports during three phases of European integration, and examined the simultaneous changes in their external competitiveness. According to Yeats (1998), trade diversion is likely to be a problem if many of the industries with rapidly increasing regional orientation record weak or diminishing external competitiveness. The argument is that these industries would hardly be able increase their regional market shares unless they were favored by regional trade preferences that give them a strategic advantage in the competition with stronger outside producers. For the case of MERCOSUR, Yeats concluded that such cases were indeed common, and that this should be interpreted as evidence of substantial trade diversion. Our analysis suggests a more unclear picture.

Looking at the first phase of European integration, the period 1962-1973, we also find that several of the industries with high growth in regional orientation record diminishing external competitiveness. Many of these industries belong to the group labelled "Land intensive", which includes agricultural products and other raw materials that were protected by high tariff and non-tariff barriers. For these product categories, it is likely that the worries expressed by Yeats (1998) are well founded, and that Europe's Common Agricultural Policy did indeed favor regional producers at the expense of more efficient producers in the rest of the world. The signs of trade diversion are somewhat weaker for the second phase of European integration, 1974-1986, with the same trend continuing to the third phase, 1987-2000. Fewer of the industries with rapidly increasing regional orientation have low and diminishing external competitiveness. Moreover, there are fewer agricultural products and raw materials with high tariff and non-tariff barriers among these industries, and more advanced industrial goods with low nominal protection. Hence, it is likely that the increases in regional sales have not been related mainly to trade preferences, but rather to other aspects of regional integration, such as increased competition and factor mobility.

One reason for this change in the impact of European integration could be the development of the multilateral trading system, together with the great increase in preferential trade agreements between the EU and its various trade partners. As a result, Europe's nominal protection levels have fallen dramatically, and most developing countries have duty-free access to a large

share of the European market. In the manufacturing sector, the average tariff has fallen below three percent, which means that there is in principle free trade in most of the product groups. Hence, the scope for trade diversion may have been reduced simply because most outsiders can enter the EU market at or near zero tariffs. As seen in Section 3, developing countries in Africa, Latin America, and Asia have apparently been able to exploit these opportunities, and their shares of EU imports have increased substantially since the early 1980s. This is another indication that trade diversion has been less important during the latest stage of EU integration than earlier.

It is also notable that the character of European integration has changed over time. The first stage of integration, with the EEC customs union as its centerpiece, could be seen as a typical example of "traditional" regional integration, where the expected trade effects are related to some reallocation of production between the member countries. The latest stage of integration, by contrast, is an example of a "modern" RIA where the main effects emanate from better use of scale economies, with higher growth, efficiency, and external competitiveness as the main results. One reason why there are weaker signs of trade diversion during this latest stage of EU integration may simply be that the concept is not well suited for the analysis of modern integration agreements. Trade diversion is said to occur when outside producers lose markets to insiders because of the establishment of a RIA. The underlying assumption is that the integration agreement does not affect the inherent production efficiency and competitiveness of firms inside and outside the RIA border, but that all effects operate through changes in relative trade costs. However, if the RIA raises the level of competition and reduces the X-inefficiency of insider firms, they may be able to capture market shares without generating any trade diversion. The most common empirical measures of trade diversion, including the one used in this study, are not well suited to situations where regional integration has this type of dynamic effects. Therefore, it is an obvious challenge for future studies to define a dynamic equivalent to the static "trade diversion". In the meantime, it is reasonable to conclude that different RIAs are likely to have different trade effects and that different stages of any specific RIA are also likely to generate different effects. From an EU perspective, it can be argued that studies of the trade effects of EU's Eastern Enlargement in 2004 are likely to be particularly interesting, not only because it is the largest of EU's expansions, but also because it is different in character to earlier expansions. All of the new member countries have income levels and factor price ratios that are markedly different from those of the old EU members, which may generate trade and investment effects that are at least qualitatively different from those at earlier stages of EU integration.

6 REFERENCES

Adams, R., P. Dee, J. Gali and G. McGuire (2003), "Trade and Investment Effects of Preferential Trade Agreements – Old and New Evidence", Productivity Commission Staff Working Paper, Canberra, May.

Aitken, N.D. (1973), "The Effect of the EEC and EFTA on European Trade: A Temporal Cross-Section Analysis", *American Economic Review,* Vol. 63, No. 5, pp. 881–892

Balassa, B. (1975), *European Economic Integration,* North Holland, Amsterdam.

Balassa, B. (1987), "Economic Integration". In *The New Palgrave: A Dictionary of Economics,* ed. John Eatwell, Macmillan, London, Vol. 2, pp. 43–47

Baldwin, R., R. Forslid and J. Haaland (1994), ""Investment Diversion and Investment Creation in Europe", *World Economy,* Vol. 19, No. 6, 635–659.

Bayoumi, T. and B. Eichengreen (1995), "Is Regionalism Simply a Diversion: Evidence from the Evolution of the EC and EFTA", CEPR Discussion Paper No. 1294.

Blomström M. and A. Kokko (1997), "Regional Integration and Foreign Direct Investment: A Conceptual Framework and Three Cases", NBER Working Paper No. 6019.

Brown, D., A. Deardorff and R. Stern (2003), "Multilateral, Regional and Bilateral Trade Policy Options for the United States and Japan", *World Economy,* Vol. 26, No. 6, pp 803–828.

Cernat, L. (2001), "Assessing Regional Trade Arrangements: Are South-South RTAs More Trade Diverting", Policy Issues in International Trade and Commodities Study Series No. 16, UNCTAD, Geneva.

Cohen, I., "MERCOSUR under Attack", Support Unit for Regional Integration Negotiations (Unidade de Apoio às Negociações de Integração Regional - UNIR). Available [online]: www.mre.gov.br/unir/webunir/BILA/ESP2/artigo5.html

Devlin, R. (1996), "In Defence of MERCOSUR". Integration, Trade and Hemisphere Issues Division, Inter-American Development Bank. Available [online]: www.iadb.org/int/itd

Devlin, R., Estevadeordal, A., and Garay, L. J. (1999), "The FTAA: Some Longer Term Issues". Integration, Trade and Hemisphere Issues Division, Inter-American Development Bank. Available [online]: www.iadb.org/int/itd

Diao, X. and Somwaru, A. (2000), "An Inquiry on General Equilibrium Effects of MERCOSUR-An Intertemporal World Model". *Journal of Policy* Modeling Vol. 22, Issue 5, pp. 557–588

Emerson, M. *et al*. (1988), *The Economics of 1992: The EC Commission's Assessment of the Economic Effects of Completing the Internal Market,* Oxford University Press, Oxford.

Estevadeordal, A., Goto, J., and Saez, R., (2000), "The New Regionalism in the Americas: The Case of MERCOSUR". Integration, Trade and Hemispheric Issues Division, Inter-American Development Bank. Available [online]: www.iadb.org/int/itd

Feenstra, R.C., R.E. Lipsey, Deng Haiyan, A.C. Ma and Mo Hengyong (2005), "World Trade Flows: 1962-2000", NBER Working Paper No. W11040.

Foroutan, F. (1998), *Does Membership in a Regional Preferential Trade Arrangement Make a Country More or Less Protectionist?* World Bank Working Paper No. 1898. Available [online]: www.mre.gov.br/unir/webunir/BILA/ESP2/artigo4.htm

Forslid, R., J. Haaland and K.H. Midelfart-Knarvik (2002), "A U-Shaped Europe? A Simulation Study of Industrial Location", *Journal of International Economics,* Vol. 57, No. 2, pp. 273–297.

Greenaway, D. and C.R. Milner (1989), "The Growth and Significance of Intra-Industry Trade", in J. Black and A. MacBean (eds), *Causes of Changes in the Structure of International Trade,* 1960–85, Macmillan, London.

Haaland, J. and V. Norman (1992), "Global Production Effects of European Integration", in L.A. Winters, ed., *Trade Flows and Trade Policy after ´1992',* Cambridge University Press, Cambridge.

Harrison, G., T. Rutherford and D. Tarr (1994), "Product Standards, Imperfect Competition, and Completion of the Market in the European Union", World Bank Working Paper No. 1293.

Harrison, G., T. Rutherford and D. Tarr (1997), "Economic Implications for Turkey of a Customs Union with the European Union", *European Economic Review,* Vol. 41, No. 3-5, pp. 861–870.

Harrison, G., T. Rutherford and D. Tarr (2002, "Trade Policy Options for Chile: The Importance of Market Access", *World Bank Economic Review,* Vol. 16, No. 1, pp. 49–79.

Harrison, G., T. Rutherford and D. Tarr (2003), "Rules of Thumb for Evaluating Preferential Trading Arrangements: Evidence from Computable General Equilibrium Assessments", *Cuadernos de Economía,* Vol. 40, No. 121, pp. 460–468.

Hoen, A.R., and Oosterhaven, J., (2004), "On the measurement of comparative advantage", University of Groningen, Research Institute SOM, Research Report O4C11.

Hudgins, E.L. (1997), "MERCOSUR Hasn't Caused Trade Diversion". *The Wall Street Journal and Cato Institute,* March 25. Available [online]: www.cato.org/dailys/3-25-97.html

Kandogan, Y. (2005), "Trade Creation and Trade Diversion Effects of Europe's Regional Liberalization Agreements", William Davidson Institute Working Paper No. 746, University of Michigan.

Krueger, A.O. (1999), "Trade Creation and Trade Diversion under NAFTA", NBER Working Paper No. 7429.

Krugman, P.R, and Obstfeld, M. (2000), *International Economics: Theory and Policy.* Addison-Wesley

Lipsey, G.R. (1957), "The Theory of Customs Unions, Trade Diversion, and Welfare", *Economica*. Vol. 24, No. 93, pp. 40–46.

Meade, J.E. (1955), *The Theory of Customs Unions,* North Holland, Amsterdam.

Mendoza, M.R., "Which MERCOSUR Anyway?", Support Unit for Regional Integration Negotiations (Unidade de Apoio às Negociações de Integração Regional – UNIR).

Nagarajan, N. (1998), *MERCOSUR and Trade Diversion: What Do Import Figures Tell Us?,* European Commission, Economic Paper No. 129

OECD (2002), *International Trade by Commodity Statistics,* OECD, Paris.

Panagariya, A. (2002 “EU Preferential Trade Arrangements and Developing Countries,” *World Economy,* Vol. 25, No. 10, pp. 1415–32.

Rutherford, T., E. Rutström and D. Tarr (1997), ”Morocco’s Free Trade Agreement with the EU: A Quantitative Assessment”, *Economic Modeling,* Vol, 14, No. 2, pp. 237–269.

Scollay, R. and J. Gilbert (2000), “Measuring the Gains from APEC Trade Liberalisation: An Overview of NGE Assessments”, *World Economy,* Vol. 23, No. 3, pp. 175–197.

Sepulveda, C. and Aguirre, A.V. (1997), “MERCOSUR: Achievements and Challenges”, Inter-American Development Bank, Working Paper Series 222

Soloaga, I., and Winters, A.L. (1999), “Regionalism in the Nineties: What Effect on Trade?”, World Bank Discussion Paper Series No. 2183.

Olarreaga, M. and Soloaga, I. (1998), “Endogenous Tariff Formation: The Case of MERCOSUR”, *The World Bank Economic Review,* Vol. 12, No. 2, pp. 297–320

Viner, J. (1950), The Customs Union Issue, Carnegie Endowment, New York.

World Bank (2000), *Trade Blocs.* A World Bank Policy Research Report, World Bank, Washington, D.C.

WTO (1995), *Regionalism and the World Trading System.* World Trade Organisation, Geneva.

Yeats, A.J. (1998), “Does MERCOSUR’s Trade Performance Raise Concerns about the Effects of Regional Trade Arrangements?”. *The World Bank Economic Review,* Vol. 12, No. 1, pp. 1–28

APPENDIX TABLE 1. COUNTRY GROUPS

EU9

Belgium, Luxembourg, Denmark, France, Germany, the Netherlands, Ireland, Italy and the UK

EU

1957–1973:
Belgium, Luxembourg, France, Germany, Italy and the Netherlands
1973–1981:
Belgium, Luxembourg, Denmark, France, Germany, Ireland, Italy, the Netherlands and the UK
1981–1986:
Belgium, Luxembourg, Denmark, France, Germany, Ireland, Italy, Greece, the Netherlands and the UK
1986–1995:
Belgium, Luxembourg, Denmark, France, Germany, Ireland, Italy, Greece, the Netherlands, Portugal, Spain and the UK
1995–2000:
Austria, Belgium, Luxembourg, Denmark, Finland, France, Germany, Greece, Ireland, Italy, the Netherlands, Portugal, Spain, Sweden and the UK

Europe

Albania, Andorra, Austria, Belgium, Bulgaria, Bosnia and Herzegovina, Bouvet Islands, Switzerland, Former Czechoslovakia, Cyprus, Czech Republic, Germany, Denmark, Spain, Finland, France, Faeroe Islands, United Kingdom, Gibraltar, Greece, Croatia, Hungary, Ireland, Iceland, Italy, Liechtenstein, Luxembourg, Monaco, Former Yugoslav Republic of Macedonia, Malta, Norfolk Island, Netherlands, Norway, Poland, Portugal, Romania, San Marino, Slovakia, Slovenia, Sweden, Vatican City State, Serbia and Montenegro, Europe n.e.s., Former Yugoslavia, Svalbard and Jan Mayen Islands, Estonia, Latvia, Lithuania, Belarus, Ukraine, Russia, Moldova, Former Soviet Union

Eastern Europe

Bulgaria, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Poland, Romania, Slovakia, Slovenia, Former Czechoslovakia

Africa

Algeria, Djibouti, Madagascar, Seychelles, Angola, Egypt, Malawi, Sierra Leone, Benin, Equatorial Guinea, Mali, Somalia, Botswana, Eritrea,

Mauritania, South Africa, Burkina Faso, Ethiopia, Mauritius, Sudan, Burundi, Gabon, Morocco, Swaziland, Cameroon, Gambia, Mozambique, Tanzania, Cape Verde, Ghana, Namibia, Togo, Central African Republic, Guinea, Niger, Tunisia, Chad, Guinea, Bissau, Nigeria, Uganda Comoros, Kenya, Réunion, Western Sahara (Sahrawi Rep), Congo, Lesotho, Rwanda, Zambia, Côte d'Ivoire, Liberia, Sao Tome and Principe, Zimbabwe, Democratic Republic of The Congo, Libya, Senegal

Asia

Afganistan, United Arab Emirates, Armenia, Azerbaijan, Bangladesh, Bahrain, Brunei Darussalam, Bhutan, Former Burma, China, Georgia, Hong Kong China, Indonesia, India, Iran, Iraq, Israel, Jordan, Japan, Kazakstan, Kyrgyzstan, Cambodia, Korea, Kuwait, Lao PDR, Lebanon, Sri Lanka, Macau, Maldives, Myanmar (ex Burma), Mongolia, Malaysia, Nepal, Neutral zone, Oman, Pakistan, Philippines, Papua New Guinea, Korea Democratic People's Republic of, Qatar, Saudia Arabia, Singapore, Syrian Arab Republic, Thailand, Tajikistan, Turkey, Turkmenistan, East Timor, Chinese taipei, Uzbekistan, Vietnam, Palestine Gaza Strip, West Indies, Asian n.e.s., Yemen

Latin America

Anguilla, Antigua & Barbuda, Argentina, Aruba, Bahamas, Barbados, Belize, Bolivia, Brazil, Chile, Colombia, Costa Rica, Dominica, Dominican Republic, Ecuador, El Salvador, French Guiana, Grenada, Guatemala, Guyana, Haiti, Honduras, Jamaica, Mexico, Netherlands Antilles, Nicaragua, Panama, Paraguay, Peru, St. Kitts, St. Lucia, St. Vincent & the Grenadines, Suriname, Trinidad & Tobago, Turks & Caicos, Uruguay, Venezuela

Oceania

American Samoa, French Southern Territories, Australia, Cocos (Keeling) Islands, Cook Islands, Christmas Island (Australia), Fiji, Micronesia, federated States of Guam, Heard Island and McDonald Islands, Kiribati, Marshall Islands, Northern Mariana Islands, New Caledonia, Niue, Nauru, New Zealand, Pacific Islands, Pitcairn Island, Palau, French Polynesia, Solomon Islands, Tokelau Islands, Tonga, Tuvalu, United States Minor Outlaying Islands, Vanuatu, Wallis and Fortuna islands, Samoa, Australian Oceania, New Zealand Oceania, Oceania n.e.s.

APPENDIX TABLE 2. INDUSTRY CATEGORIES

SITC Rev. 2	Industry	Industry class
0	Live animals chiefly for food	Land intensive
1	Meat and meat preparations	Land intensive
2	Dairy products and birds' eggs	Land intensive
3	Fish, crustaceans, mollucs, preparations thereof	Land intensive
4	Cereals and cereal preparations	Land intensive
5	Vegetables and fruit	Land intensive
6	Sugar, sugar preparations and honey	Land intensive
7	Coffee, tea, cocoa, spices, manufactures thereof	Land intensive
8	Feeding stuff for animals, not including unmilled cereals	Land intensive
9	Miscellaneous edible products and preparations	Land intensive
11	Beverages	Land intensive
12	Tobacco and tobacco manufactures	Land intensive
21	Hides, skins and furskins, raw	Land intensive
22	Oil seeds and oleaginous fruit	Land intensive
23	Crude rubber (including synthetic and reclaimed)	Land intensive
24	Cork and wood	Land intensive
25	Pulp and waste paper	Land intensive
26	Textile fibers (except wool tops) and their wastes	Land intensive
27	Crude fertilizers and crude materials (excluding coal)	Land intensive
28	Metalliferous ores and metal scrap	Land intensive
29	Crude animal and vegetable materials, n.e.s.	Land intensive
41	Animal oils and fats	Land intensive
42	Fixed vegetable oils and fats	Land intensive
43	Animal-vegetable oils-fats, processed, and waxes	Land intensive
94	Animals, live, zoo animals, dogs, cats etc.	Land intensive
53	Dyeing, tanning and coloring materials	Human capital intensive
55	Essential oils & perfume materials; toilet and cleansing preparations	Human capital intensive
76	Telecommunications & sound recording apparatus	Human capital intensive
78	Road vehicles (including air-cushion vehicles)	Human capital intensive

79	Other transport equipment	Human capital intensive
88	Photographic apparatus, optical goods, watches	Human capital intensive
61	Leather manufactures, n.e.s	Capital intensive
62	Rubber manufactures, n.e.s.	Capital intensive
63	Cork and wood manufactures (excluding furniture)	Capital intensive
64	Paper, paperboard, articles of paper, paper-pulp/board	Capital intensive
65	Textile yarn, fabrics, made-up articles, related products	Capital intensive
66	Non-metallic mineral manufactures, n.e.s.	Capital intensive
67	Iron and steel	Capital intensive
68	Non-ferrous metals	Capital intensive
69	Manufactures of metal, n.e.s.	Capital intensive
81	Sanitary, plumbing, heating and lighting fixtures	Labour intensive
82	Furniture and parts thereof	Labour intensive
83	Travel goods, handbags and similar containers	Labour intensive
84	Articles of apparel and clothing accessories	Labour intensive
85	Footwear	Labour intensive
89	Miscellaneous manufactured articles, n.e.s.	Labour intensive
51	Organic chemicals	Technology intensive
52	Inorganic chemicals	Technology intensive
54	Medicinal and pharmaceutical products	Technology intensive
56	Fertilizers, manufactured	Technology intensive
57	Explosives and pyrotechnic products	Technology intensive
58	Artificial resins, plastic materials, cellulose esters and ethers	Technology intensive
59	Chemical materials and products, n.e.s.	Technology intensive
71	Power generating machinery and equipment	Technology intensive
72	Machinery specialized for particular industries	Technology intensive
73	Metalworking machinery	Technology intensive
74	General industrial machinery & equipment, and parts	Technology intensive
75	Office machines & automatic data processing equipment	Technology intensive
77	Electrical machinery, apparatus & appliances n.e.s.	Technology intensive
87	Professional, scientific & controlling instruments	Technology intensive

95	Arms, of war and ammunition therefor	Technology intensive
32	Coal, coke and briquettes	Energy intensive
33	Petroleum, petroleum products and related materials	Energy intensive
34	Gas, natural and manufactured	Energy intensive
35	Electric current	Energy intensive

SAMMANFATTNING PÅ SVENSKA

EUROPEISK INTEGRATION OCH HANDELSOMFÖRDELNING

Ari Kokko, Thomas Mathä och Patrik Gustavsson Tingvall

De senaste årtiondena har inte enbart karakteriserats av en allt djupare regional integrationsprocess i Europa, utan regional integration har varit ett centralt begrepp på de flesta andra håll i världen. Effekterna av regionala avtal har också diskuterats intensivt, inte minst i Latinamerika, där den regionala tullunionen MERCOSUR bidrog till mycket stora ökningar av den bilaterala handeln mellan medlemsländerna under första halvan av 1990-talet. Även om denna utveckling tolkades positivt på de flesta håll förekom det också skarp kritik. En av de mest uppmärksammade kritikerna var Alexander Yeats vid Världsbanken, som 1998 publicerade en mycket omtalad rapport där han hävdade att MERCOSUR bidragit till omfattande handelsomfördelning och välfärdsförluster bland MERCOSURs handelspartners. Detta påstående stöddes av en detaljerad analys av MERCOSURs interna handel på branschnivå. Den snabbaste ökningen i den interna handeln återfanns i produktgrupper med låg eller sjunkande extern konkurrenskraft. Enligt Yeats kunde detta mönster endast uppkomma om de relativt svaga regionala producenterna gynnades av diskriminerande handelsregler som hjälpte dem att konkurrera ut mera effektiva producenter från andra delar av världen. Mera specifikt analyserade Yeats den samtidiga utvecklingen av två index på branschnivå: det första var ett mått på handelns regionala orientering (regional orientation index – ROI) medan det andra var ett mått på branschens externa konkurrenskraft (revealed comparative advantage index – RCA).

Syftet med denna rapport är att undersöka utvecklingen av Europas interna handel med den metodologi som introducerades av Yeats (1998). Rapporten består av tre substantiella avsnitt, där det första avsnittet diskuterar Yeats resultat och kritiken mot slutsatserna. Den allvarligaste kritiken gäller den bristande analysen av MERCOSURs import. För att omvärlden ska missgynnas allvarligt av handelsomfördelning krävs att omvärldens exportmöjligheter försvagas på grund av de privilegier som ges åt regionala producenter. Några sådana tecken går dock inte att finna: MERCOSURs import ökade kraftigt även i de produktgrupper som registrerade den högsta ökningen i regional orientering. En annan brist gäller kriterierna för urvalet av industrier med stor ökning av den regionala orienteringen. Yeats valde de 30 produktgrupper som uppvisade den största procentuella ökningen i ROI, men dessa produktgrupper var relativt små och stod endast för en liten del av den totala tillväxten i regional handel. Ett urval med fokus på de produktgrupper som registrerat de största absoluta ökningarna av den

regionala handeln uppvisade till exempel inte någon svag eller minskande extern konkurrenskraft, utan snarare motsatsen.

Det andra avsnittet utgörs av en genomgång av de främsta stegen i den europeiska integrationsprocessen, samt en sammanfattning av Europas handelsstruktur. Rapporten skiljer mellan tre faser av integration. Den första omfattar perioden från 1957, då den europeiska tullunionen EEC bildades, till 1973, då Storbritannien, Irland och Danmark anslöt till avtalet. Den andra perioden utgörs av åren 1974-1986, och kan ses som en mellanfas av integrationsprocessen. Även om det europeiska samarbetet fördjupades genom EMS, det europeiska monetära systemet, och Greklands, Portugals och Spaniens medlemskap under 1980-talet, hände inte mycket på handelsfronten. Den tredje och sista perioden sträcker sig mellan 1987 och 2000, och karakteriseras av en omfattande fördjupning av samarbetet inom ramarna för den inre marknaden (the Single Market). Medan den första fasen kan ses som en illustration av ett klassiskt integrationsavtal, där tullunionen syftar till att skapa en mer effektiv fördelning av produktionen inom regionen, utgör den inre marknaden ett exempel på modern integration, där de främsta effekterna är relaterade till tillväxt och konkurrenskraft. Genom att ta bort alla hinder mellan de nationella marknaderna avsåg man öka konkurrensen och tvinga fram en koncentration av industristrukturen, så att de överlevande företagen skulle kunna växa sig större och starkare. Därmed skulle de också generera snabbare tillväxt och högre inkomster.

Det andra avsnittet beskriver också EUs relationer med utvecklingsländer och transitionsekonomier. Som ett resultat av en mängd olika bilaterala handelsavtal har EU utsträckt olika tullförmåner till de allra flesta länder. Ett knappt 80-tal u-länder deltar i Cotonou-avtalet, som omfattar både bistånd och handelspreferenser. De 50 fattigaste länderna har ett avtal med rubriken Everything But Arms, som ger tullfritt tillträde till den gemensamma marknaden för alla varor utom vapen och ammunition (och, under en övergångstid, bananer, ris och socker). De europeiska transitionsekonomierna slöt associationsavtal med EU i mitten av 1990-talet, för att senare ansluta som fullvärdiga medlemmar. Medelhavsländerna har särskilda avtal som syftar till att skapa ett frihandelsområde senast 2010. Faktum är att endast sex länder – USA, Kanada, Japan, Australien, Nya Zeeland och Taiwan – saknar särskilda avtal som ger bättre villkor vid export till Europa än GATTs mest-gynnad-nationsklausul. Ett resultat av dessa handelsförmåner är att både Afrika, Latinamerika, Asien, och Europas transitionsekonomier ökat sina andelar av EUs import sedan början av 1980-talet.

Det tredje avsnittet undersöker data för europeisk export och import på branschnivå för att identifiera de industrier som kan tänkas påverkas av

handelsomfördelning. Analysen görs separat för de tre olika faserna av europeisk integration. Resultaten för den första perioden, 1962-1973, ger visst stöd för farhågorna gällande handelsomfördelning. De flesta produktkategorierna som uppvisar snabbt ökande regional orientering är råvaror och jordbruksprodukter med relativt höga skyddsnivåer och låg extern konkurrenskraft. Det är troligt att den regionala handeln inom dessa varugrupper ökade på bekostnad av import från mera effektiva producenter utanför EU, som missgynnades av handelsomfördelningen. Även den andra fasen av regional integration, 1973-1986, uppvisar vissa tecken på handelsomfördelning, men i mindre utsträckning än den föregående perioden. Den tredje fasen, som bland annat omfattar utvecklingen av den inre marknaden, skiljer sig däremot från det tidigare mönstret. De största ökningarna i den interna EU-handeln registreras för relativt avancerade industrivaror med låga nominella tullnivåer, där EU dessutom uppvisar relativt hög extern konkurrenskraft: det är osannolikt att handelsomfördelning är ett stort problem för dessa varugrupper. Även andra studier verkar stödja denna slutsats: tidiga analyser av det europeiska samarbetet pekar ofta på omfattande handelsomfördelning, medan senare studier inte finner starka spår av motsvarande snedvridningar. Utrymmet för handelsdiskriminering har helt enkelt minskat i takt med att skyddsnivåerna för den europeiska importen har minskat, både som ett resultat av förhandlingarna inom ramarna för GATT och WTO och genom de många preferentiella handelsavtal som EU slutit med sina grannar och största delen av den tredje världen. En av slutsatserna från analysen är således att handelseffekterna av regional integration kan variera både mellan olika integrationsavtal och över tiden inom en och samma region, i takt med att den internationella omgivningen förändras.

Förändringen i handelsmönstret är sannolikt också relaterad till skiftet i den europeiska integrationsprocessen, från traditionell tullunion till en djup integrationsprocess med fokus på marknadsintegration, konkurrens och skalfördelar. Medan den traditionella regionala integrationen sökte fördelar från en mera rationell regional fördelning av produktionen, men antog att teknologin och effektiviteten i den regionala produktionen förblev oförändrad, har den ”nya” integrationen som mål att förändra de inhemska företagens konkurrenskraft. Om detta mål uppnås kommer regionala företag att lyckas ta marknadsandelar från utländska konkurrenter på den regionala marknaden eller på andra marknader, beroende på utgångsläget. Den förändring i handelsmönstret som då uppkommer kan missgynna producenter i andra delar av världen, men kan knappast karakteriseras som klassisk handelsomfördelning, eftersom orsaken till att marknadsandelarna förändras är regionens stärkta konkurrenskraft. De empiriska mått på handelsomfördelning som oftast används i forskningen – inklusive det mått som diskuterats

i denna rapport – kan dock inte särskilja mellan ”sann” handelsomfördelning som beror på diskriminerande handelsregleringar och förändringar som grundas på stärkt konkurrenskraft. En av utmaningarna för den empiriska forskningen om integrationens handelseffekter är därför att söka nya definitioner av och mått på handelsomfördelning i ett dynamiskt perspektiv.

SIEPS PUBLICATIONS AVAILABLE IN ENGLISH

2005:5
From Policy Takers to Policy Makers:
Adapting EU Cohesion Policy to the Needs of the New Member States
Editors: Jonas Eriksson, Bengt O. Karlsson and Daniel Tarschys

2005:4
The Enigma of European Added Value:
Setting Priorities for the European Union
Author: Daniel Tarschys

2005:1op
The 2005 Luxembourg Presidency:
A Presidency Devoted to the Stability and Growth Pact
and to the Lisbon Process
Authors: Patrick Dumont and Philippe Poirier

2004:9 *The Political Dynamics of Turkish Accession to the EU:*
A European Success Story or the EU's Most Contested Enlargement?
Author: Kirsty Hughes

2004:1op
The Netherlands 2004 EU Council Presidency
– Dutch EU Policy-making in the Spotlights
Author: Mendeltje van Keulen

2004:1u
European Governance
– an Overview of the Commission's Agenda for Reform
Authors: Josefin Almer and Matilda Rotkirch

2003:1op
Contrasting Transatlantic Interpretations
– the EU and the US towards a Common Global Role
Author: Ludger Kühnhardt

2003:19
Industrial Structure and Industry Location in an Enlarged Europe
Author: Karolina Ekholm

2003:18
Coming of Age? Economic Management of the European Union
Authors: Per Molander and Allan Gustafsson

2003:17
Reinventing Cohesion: The Future of European Structural Policy
Author: Daniel Tarschys

2003:14
Decentralized Agencies and the IGC – a Question of Accountability
Authors: Carl Fredrik Bergström and Matilda Rotkirch

2003:9
Reforming the Council – a Work in Progress
Authors: Helen Wallace and Fiona Hayes-Renshaw

2003:8
Simply Simplification? The Proposal for a Hierarchy of Legal Acts
Authors: Carl Fredrik Bergström and Matilda Rotkirch

2003:7
The Invisible Transformation of Codecision
– Problems of Democratic Legitimacy
Authors: Henry Farrell and Adrienne Héritier

2003:1
The Open Method of Coordination
– a New Governance Architecture for the European Union?
Author: Claudio M Radaelli



Swedish Institute for European Policy Studies

Universitetsvägen 10 F
SE-106 91 Stockholm
Office: Stockholms universitet,
Frescati, House F, 6th floor
Tel: +46-(0)8-16 46 00
Fax: +46-(0)8-16 46 66
E-mail: info@sieps.se
www.sieps.se